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AN ANALYSIS OF IMPACT OF THE MAGADI ROAD ON THE URBAN SPATIAL OF THE PATTERN AND PROVISION OF SERVICES

A CASE STUDY OF MAGADI ROAD ONGATA RONGAI TOWN, KAJIADO COUNTY

BY:

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DECLARATION

STUDENT DECLARATION

I **CLEVARENCE OTIENO OSANO** hereby declare that this research project is my original work and has not been submitted to any other university or institution of higher learning for academic award.

SIGNED………………….........………….DATE……………………………………

SUPERVISER DECLARATION

This research project has been submitted for examination with the approval as per the University supervisor.

PROF. JEREMIAH AYONGA

SIGNED………………………………DATE…………………………………………

DEDICATION

I would like to dedicate this Research Project to my late grandfather Boaz Ogweno Odiyo and my father Erustus Osano Ogweno who always reminded to continue pushing on no matter the circumstances, because there is always light at the end of the tunnel. Through their words of aspiration as made me to reach the level I am right now even after there had been struggle and felt like giving up.

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I wish to appreciate the Almighty God for providing me with the good health and resource to undertake this academic work. Am very grateful to my supervisor Prof. Jeremiah Ayonga for his continued constructive contribution towards the pursuance of the research, I wish to point out the fact that without his support and encouragement, this work would have not been accomplished to this level.

I would like to recognize the role of Dr Elizabeth, Dr Musyoka and Dr Opiyo the course coordinators for their constructive contribution during research period. I also wish to appreciate my classmates for their support and encouragement. Lastly, I would like to appreciate my entire family for their support and encouragement.

ABSTRACT

This study investigates into the impact of Magadi Road on the urban spatial pattern and service provision in Ongata Rongai Town, Kajiado County. Firstly, it investigates the key role of transport infrastructure, specifically Magadi Road, in shaping the linear urban pattern observed in the area. Through observed analysis and spatial modelling, the study aims to unveil the complex connections between transportation networks and urban development, shedding light on how Magadi Road has influenced settlement patterns along its trajectory.

Secondly, the research inspects the consequences of this linear urban pattern on the provision of essential services and utilities. By examining accessibility, distribution, and quality of services such as water, electricity, healthcare, and education along Magadi Road, the study aims to uncover disparities and challenges in service delivery attributable to the linear spatial layout.

Furthermore, the socioeconomic outcomes of the linear pattern are thoroughly examined. This includes analysing factors such as employment opportunities, income levels, social cohesion, and overall community well-being along the Magadi Road corridor. By understanding the social and economic dynamics influenced by the linear urban pattern, the study provides insights into both its benefits and drawbacks for residents and businesses.

Lastly, the research proposes mitigation strategies to address the challenges arising from the linear pattern identified throughout the study. Through a combination of urban planning interventions, policy recommendations, and community engagement initiatives, the study seeks to offer practical solutions to enhance the liveability, sustainability, and resilience of Ongata Rongai Town and its surrounding areas affected by the linear urban development along Magadi Road.

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CHAPTER ONE: INTRODUCTION

1.0 Background of the study Area

In the dynamic landscape of urban development, the role of road infrastructure is fundamental in shaping the spatial patterns of towns or cities. As urban areas expand and evolve, the layout and design of road networks become critical factors influencing the organization and accessibility of different regions within a town or city. This analysis seeks to explore the impact of road infrastructure on the spatial fabric of urban environments, investigating into the ways in which road systems contribute to the formation of distinct land use patterns, connectivity, and overall urban structure.

The relationship between road infrastructure and urban spatial patterns is a subject of increasing importance, as cities worldwide struggle with the challenges of efficient transportation, sustainable development, and the accommodation of growing populations. The design and distribution of roads not only affect the physical flow of traffic but also play a significant role in determining the economic, social, and environmental aspects of urban life. By understanding how road networks influence the spatial organization of cities, us urban planners and policymakers can make informed decisions to create more liveable, accessible, and resilient urban environments. This sets the stage for the research to explore, analyse and propose solutions for the critical problems identified.

1.1 Problem Statement

Ongata Rongai, a rapidly growing urban area, has witnessed a distinctive linear urban pattern along Magadi Road. This pattern is primarily attributed to the influence of road infrastructure. However, the implications of this spatial configuration on the overall urban fabric, service provision, utilities, and socioeconomic conditions remain poorly understood. The existing linear pattern, shaped by road infrastructure, raises critical questions regarding its impact on the community's development, functionality, and well-being. Therefore, there is need for an inclusive analysis to investigate the role of transport infrastructure in creating this linear urban pattern along Magadi Road and to assess its effects on service delivery, utilities, and the socioeconomic dynamics of Ongata Rongai. Additionally, identifying practical mitigation strategies is imperative to address challenges arising from this unique spatial arrangement and ensure the sustainable growth and resilience of the urban community.

1.2 RESEARCH QUESTIONS

1. What is the role of transport infrastructure in creating the linear urban pattern along Magadi road?
2. How does the linear pattern affect the provision of services and utilities?
3. What are the socioeconomic outcomes of the linear pattern?

4. What planning framework can mitigate the challenges arising from the linear pattern?

1.2 OBJECTIVES

1. **To investigate the role of transport infrastructure in creating the linear urban pattern along Magadi road.**

2. To examine the effects of the linear pattern on provision of services and utilities.

3. To examine the socioeconomic outcomes of the linear pattern.

4. To provide mitigation strategies to address the challenges arising from the linear pattern.

1.4 JUSTIFICATION

Ongata Rongai, located on the outskirts of Nairobi, Kenya, has witnessed significant changes in its urban landscape, particularly along Magadi Road, where a distinct linear pattern has emerged. This transformation is closely linked to the influence of road infrastructure. Understanding the role of transport infrastructure in shaping this unique urban pattern is crucial for deciphering the factors behind the city's spatial arrangement. By selecting Ongata Rongai as the case study, this research aims to unravel the intricacies of how roads contribute to the city's linear development, providing valuable insights for urban planners, policymakers, and local authorities grappling with similar growth dynamics.

The second facet of this study delves into the practical implications of the observed linear pattern, focusing on how it impacts the provision of essential services and utilities in Ongata Rongai. By assessing the effects on service delivery, accessibility, and infrastructure use, the research aims to provide actionable insights. This examination is essential as it directly connects the physical layout of the city to the daily experiences of its residents, shedding light on potential challenges and areas for improvement.

Furthermore, the research extends its scope to understand the socioeconomic outcomes of Ongata Rongai's linear urban pattern. This involves examining changes in employment, economic activities, and overall community well-being. By doing so, the study seeks to offer a comprehensive evaluation of how the city's growth patterns shape the lives of its residents. Ultimately, by providing targeted mitigation strategies to address challenges arising from this unique spatial arrangement, the research aspires to contribute not only academically but also practically, empowering local authorities and stakeholders to navigate growth challenges and foster a resilient and harmonious urban environment in Ongata Rongai.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

The chapter discusses the literature on the impact of road infrastructure on the urban spatial pattern and its theoretical framework. It's also discusses the case studies done for different towns in different countries and writings on sustainability of urban areas particularly in terms of linear spatial urban pattern. The chapter also review on the policy and legal framework. Lastly the chapter discusses on conceptual framework.

Definition and characteristics of linear urban pattern

A linear urban pattern is a distinctive spatial arrangement where urban development stretches along a linear or elongated axis. The city or settlement is organized in such a way that its primary form follows a specific corridor, typically defined by a major transportation route, like a road, river, coastline, or railway. This linear axis serves as the backbone of the city's layout, influencing the distribution of various land uses and activities.

One of the key characteristics of linear urban patterns is the concentration of development along the linear corridor. This results in a higher density of residential, commercial, and industrial activities in close proximity to the main axis. Additionally, land use zoning tends to follow a linear or strip pattern, with designated areas for specific purposes along the linear route. This organized structure contributes to spatial efficiency, maximizing land use along the defined axis.

Transportation plays a crucial role in linear urban patterns, with major transportation corridors such as highways or railways shaping the development. The linear nature of these patterns enhances accessibility, facilitating the movement of people and goods along the corridor. The concentration of activities along the linear axis often leads to the establishment of central points, serving as focal hubs for civic, cultural, or commercial functions.

While linear urban patterns offer advantages in terms of spatial efficiency and accessibility, they also pose challenges. Expansion opportunities may be limited perpendicular to the linear axis, and congestion issues can arise along the main transportation routes. Nevertheless, the specific characteristics of linear urban patterns can vary based on the historical, geographical, and economic factors influencing urban development in a particular region. Examples of linear urban patterns can be observed in cities worldwide, each shaped by its unique context and influences.

2.1 Causes of linear urban pattern

The linear urban pattern observed in many cities can often be traced back to a combination of historical, geographical, and economic factors. One significant influence is the presence of transportation routes, such as rivers, coastlines, roads, or railways, along which settlements tend to develop for the convenience and accessibility they offer in terms of trade and transportation. Additionally, the topography of the region plays a crucial role, with cities adapting their layout to follow natural features like valleys or ridges, creating a linear arrangement that aligns with the contours of the land. Historical factors, such as the initial settlement patterns driven by trade routes, defensive considerations, or the location of resources, contribute to the organic growth of cities, often resulting in linear developments. Zoning regulations and land-use policies, reflecting economic activities and resource distribution, can further shape the linear urban pattern. Infrastructure development, including highways, railways, or canals, plays a vital role, as does intentional government planning efforts aimed at promoting efficient land use and transportation connectivity. Cultural and social preferences, along with environmental considerations, also contribute to the emergence of linear urban patterns, creating a complex interplay of factors that define the spatial organization of cities.

2.2 Effects of linear growth pattern along roads

Linear growth patterns along roads can give rise to several undesirable effects that impact the social, economic, and environmental aspects of communities.

The linear pattern often exacerbates traffic congestion along Magadi Road, impacting accessibility and mobility. This observation resonates with the works of Donald Appleyard, whose research in "Liveable Streets" emphasizes the social consequences of heavy traffic on urban communities. Appleyard highlights how congestion can disrupt neighbourhood life and hinder pedestrian mobility, underscoring the need for transportation planning that prioritizes liveability and sustainability.

Along Magadi Road, the linear pattern fosters intensified land use, characterized by a concentration of commercial, industrial, and residential developments. This phenomenon aligns with Jane Jacobs' insights in "The Death and Life of Great American Cities," where she discusses the importance of mixed-use development in creating vibrant urban environments. Jacobs emphasizes the vitality that arises from diverse land uses and active street life, illustrating how the linear pattern can contribute to urban vibrancy and economic vitality.

The linear urban pattern along Magadi Road serves as a catalyst for economic development, attracting businesses and stimulating employment opportunities. This observation echoes the principles of urban economics, as elucidated by authors such as Richard Florida in "The Rise of the Creative Class." Florida emphasizes the role of urban amenities and infrastructure in fostering innovation and economic growth, highlighting how corridors like Magadi Road can become engines of prosperity and opportunity.

Despite its economic benefits, the linear pattern may exacerbate social segregation by creating barriers between neighbourhoods and communities. This insight draws from the works of urban sociologists such as William Julius Wilson, who examines the spatial dimensions of poverty and inequality in cities. Wilson's research underscores how spatial segregation perpetuates social disparities, reflecting on the challenges faced by marginalized communities living adjacent to corridors like Magadi Road.

The linear urban pattern contributes to environmental degradation along Magadi Road, with increased air and noise pollution from vehicular traffic. This observation aligns with environmental planning theories, as expounded by authors like Ian McHarg in "Design with Nature." McHarg emphasizes the importance of ecological considerations in urban planning, advocating for strategies that minimize the environmental footprint of transportation infrastructure and promote sustainable development.

The linear pattern places strain on infrastructure and public services, necessitating substantial investments to meet growing demands. This challenge resonates with the infrastructure planning literature, with authors like Anthony Downs in "Stuck in Traffic" highlighting the funding gaps and capacity constraints facing urban infrastructure systems. Downs underscores the importance of innovative financing mechanisms and strategic investments to address infrastructure challenges along corridors like Magadi Road.

The linear pattern may fragment communities along Magadi Road, limiting social interactions and collective identity. This observation draws from the field of social geography, with authors such as Neil Smith exploring the spatial dimensions of social inequality and urban segregation. Smith's work underscores how linear corridors can act as physical barriers that divide communities, highlighting the importance of inclusive planning approaches that promote social cohesion and connectivity.

Despite its challenges, the linear urban pattern presents opportunities for transit-oriented development (TOD) along Magadi Road. This observation aligns with TOD principles articulated by urban planners like Peter Calthorpe in "The Next American Metropolis." Calthorpe emphasizes the benefits of compact, mixed-use development around transit nodes, illustrating how corridors like Magadi Road can support sustainable transportation options and foster vibrant, walkable communities.

2.3 The Origin, Growth, and Development of Road Infrastructure and its effect on urban pattern

One of the primary factors that affect developmental patterns in both urban and regional levels is road networks. In most cases, new links can lead to agglomeration, activities dispersed, and the altercation of the balance between where the households are located as well as the level of aggregation of firms but it is often dependent on the type of change to the network as written by Iacono & Levinson, 2009.

The need for movement of people and commodities has been an everlasting requirement for many centuries. These has been continuous as the need to survive has continues to persist. This has led to continuous improvement on roads network from Stone Age to the modern day (Mouratidis & Kehagia, 2014). Transportation of commodities is the apparently the main socioeconomic activity positively impacting people's welfare. Road transport is the most common and popular means of transport. The evolution of road networks can be seen through road expansion, road resurfacing, or the construction of new road networks.

Therefore, road networks and spatial urban pattern changes are assumed to have an influence over each other over time mutually. When a new road link is constructed or expanded, it would have a great influence on the choice of household location, developments in the real estate sector, land development and also the density of building which in hand seems to create a certain urban spatial pattern in terms of the arrangement. Accessibility and demand for travel would also ultimately be influenced by the changes in urban pattern due to changes in land-use.
The improvement and expansion of road networks in an urban area result in urban expansion through the changes in the existing land uses which will take specific direction depending on the direction of the road expansion.

Burchfield studied the U.S urban development and expansion and determinants of the spatial differences during expansion. Remote sensing data was used in tracking how land-use evolved monitoring the direction of pattern they take. The findings indicated that road networks increase the likelihood of urban expansion. Therefore, the expansion of road networks over time results in more accessibility and thus encourages changes in land-use patterns. Improving road networks play an important part in the urban landscape by influencing urban land use and guiding changes to the urban environment.

Ogonda (1986) examined the origin, growth, and development of systems of road networks in Kenya and how it is related with socio-economic development factors. He identified the main steps of evolution and compared them with a sequence model that was ideal for the growth of road networks and development. In his study, he assesses and evaluates show network patterns relate and the indices that have been chosen for the development of socio-economic factors in accordance with the units of administration in which those districts are classified in the different levels of development then they are related to the level of road transport development. Here a relationship was derived to determine the interrelationships between road networks and development due to various land uses which take specific urban pattern. The study findings showed that there is a close relation between development of road networks and features of socio-economic development.

2.3.1Road Infrastructure Pattern and Connectivity

One of the major primary elements of infrastructure that is known to contribute to the economic growth and provision of strength and security to countries is road networks. It consists of many interwoven roads exhibiting many patterns by Zhang & Lund University. According to Aderamo, 2003, the road network provides accessibility required by different land uses, therefore they are considered as an important element in urban development pattern. In order for such urban areas to properly function, efficient network of transport should be in place. He argues that the road network analysis should involve recognizing the qualities of the roads and how their patterns which Magadi road forms a linear pattern due to its nature of expansion.
Road infrastructure is considered a key aspect to the regional growth of an urban area. The huge cost of road development as an infrastructure demands proper utilization. That can be gotten only when the connection and orientation are proper. Consequently, organic growth pattern is more predominant in many urban areas hence the need for studies on road network connectivity pattern as connectivity level of the road network influences how land-use patterns develop in a region.

More road network connectivity tends to attract competitive land uses which in hand tend to take a particular direction depending on the availability and cost of land and this study will help determine the more dominant uses of land in the area of study and how they bring out the urban spatial pattern. (Ali, 2001) highlights that street are an indication of an areas historical background, topographical layout and economic wellbeing as they take many shapes and forms. Transportation infrastructure is a key activity as its importance can be related to human activities and road networks have to be connected to various regions to achieve the goal of movement by Land Transport Regulatory Commission 2016.

2.3.2 Roads, Urban & Economic Development and Modernization on urban pattern

African investors believed that investments in transport infrastructure positively influenced economic development even though the relationship between road networks and economic development has been contentious.

The connection between development pattern and transportation infrastructure seems obvious in most towns and cities. The belief that 'growth will be generated through public investment' has justified resource allocation to the transportation sector, encouraging economic growth and development which when observed tends to take a particular direction. European colonial authorities were equally convinced by theories connecting transportation infrastructure to economic development and transformation.

As mentioned by (Leinbach, 1975), it was generally perceived that transportation plays an important part in spreading modernization. He argued that 'the potentiality to move items and people reliably while joining supply and demand is an essential result of enhanced transportation. However, some researchers such as (Dalenberg & Partridge, 2006)have brought out proof questioning the validity of studies suggesting an optimistic relationship between road infrastructure development and economic development. They noted some road networks might negatively impact development, such as unpaved roads. As argued by (Hoyle & Smith, 1992), transportation infrastructure acts as a catalyst for social activities and as a promoter of economic development which this study try to find out how does this road infrastructure affect the pattern of thses development.

It enhances social processes by lowering the cost of movement for especially people, goods and services necessary for improving wellbeing. To understand road infrastructure as a catalyst of urban pattern and economic development, Njoh (2000) stated that the frame work of the transportation sector must be widened so as to help include all activities that may not be traditionally treated as part of the transportation industry, such as supplying construction materials & equipment operating transportation facilities and managing transportation services. He postulates that by thinking of the transport sector in this broader sense, it is easy to appreciate that the sector attracts a wider component of the economy. Any efforts made to improve the sector directly improve a region's entire economy.

2.3.3 Influence of Road Infrastructure Development on Urban Spatial Pattern

Urban Spatial pattern is a term frequently used to denote and discuss the distribution of activity within a metropolitan area. The most important surveys to be carried out by planners before the planning phase is land use studies, which includes thorough studies on land use types in an area based on land use distribution patterns so as to decide the type of land use to be located in different city parts which in hand produce a certain alignment (Al Khaldi, 2005).

Road infrastructure development plays a crucial role in shaping the spatial pattern of urban areas. The impact of such development extends beyond mere transportation, influencing the overall structure and organization of the city. This influence can be observed in various aspects, including land use, accessibility, and socio-economic dynamics.

Road infrastructure acts as a catalyst for land use changes within urban areas. The construction of new roads or the improvement of existing ones often leads to increased accessibility to different parts of the city. This accessibility, in turn, attracts various land uses such as residential, commercial, and industrial developments. As a result, urban spatial patterns evolve, with areas adjacent to well-connected roads experiencing higher levels of development and economic activity.

Moreover, the creation of an efficient road network contributes to the emergence of urban nodes and corridors. Nodes, often located at intersections or major junctions, become focal points for concentrated development. Corridors, formed along major roads, showcase linear patterns of development. These nodes and corridors form the backbone of the urban spatial structure, guiding the growth and expansion of the city.

The influence of road infrastructure on the spatial pattern is also evident in the development of urban sprawl. Improved connectivity through roads encourages the outward expansion of the city, leading to the sprawl of residential and commercial areas. This horizontal expansion can result in the formation of suburbs and peri-urban areas, altering the overall spatial dynamics of the urban landscape.

Furthermore, road infrastructure development affects the socio-economic fabric of urban areas. Well-connected regions often witness higher property values and increased economic opportunities, attracting businesses and residents. Conversely, areas with poor road infrastructure may experience economic stagnation and face challenges in attracting investments. Thus, the spatial distribution of infrastructure can exacerbate social and economic inequalities within a city.

Road infrastructure development is a key driver in shaping the spatial pattern of urban areas. It influences land use, accessibility, and socio-economic dynamics, contributing to the formation of nodes, corridors, and urban sprawl. As cities continue to grow and evolve, the strategic planning and development of road infrastructure will play a key role in determining the spatial organization and sustainability of urban areas.

2.4 Case study

LOS ANGELES CITY, USA

Los Angeles, often characterized by its vast urban expanse and diverse neighbourhoods, exhibits a distinctive urban spatial pattern shaped significantly by its road infrastructure. The city's extensive network of freeways, including iconic ones such as the 405 and 101, plays a pivotal role in defining the spatial organization. The radial design of these freeways creates a hub-and-spoke pattern, with the downtown area acting as a central node from which various corridors extend outward. This radial structure has contributed to the development of distinct urban nodes and corridors, each with its own unique character and function.

The freeway-centric layout of Los Angeles has also influenced patterns of urban development. Adjacent to major highways, there is often a concentration of commercial and business activities, forming nodes of economic vitality. On the other hand, the areas farther from these transportation arteries may exhibit different land uses, such as residential neighbourhoods or industrial zones. This spatial heterogeneity reflects the impact of road infrastructure on the distribution of urban functions.

Urban sprawl is a notable feature of Los Angeles' spatial pattern, and it is intricately linked to the city's road network. The development of numerous freeways has facilitated the outward expansion of the city, leading to the creation of suburbs and peri-urban areas. The availability of efficient road connections has played a role in fostering residential and commercial growth in these suburban regions, contributing to the sprawling nature of the Greater Los Angeles area.

Despite the advantages of a well-connected road system, Los Angeles also faces challenges related to traffic congestion and the environmental impact of extensive car use. The heavy reliance on automobiles has shaped not only the physical landscape but also the daily experiences of residents navigating the city. Efforts to address these issues, such as investments in public transportation and the promotion of alternative modes of commuting, underscore the ongoing efforts to redefine the urban spatial pattern in Los Angeles.

In conclusion, Los Angeles' urban spatial pattern is intricately tied to its road infrastructure, characterized by a radial freeway network, distinct urban nodes along major corridors, and the phenomenon of urban sprawl. The city's layout reflects both the opportunities and challenges associated with a car-centric urban model, emphasizing the ongoing need for thoughtful planning and sustainable transportation solutions.

The Car in the City

Exploring the Relationship between Roads and Urban Form" by Alan Buckingham explores into several implications that arise from the dynamic interaction between roads and urban development. Spatial Transformation-The book underscores how roads play a pivotal role in the spatial evolution of cities. The implications extend to the physical layout of urban areas, determining the location of neighbourhoods, commercial zones, and public spaces. As roads expand and adapt, they become primary drivers of the city's growth and development.

Social Dynamics- Buckingham explores how road networks influence social interactions and community dynamics. Well-designed roads can facilitate connectivity and foster a sense of community, while poorly planned infrastructure may lead to physical and social fragmentation. The book emphasizes the social implications of road design and placement on neighbourhood cohesion and overall community well-being. Cultural and Aesthetic Significance-The presence of roads is examined in terms of its contribution to the character and identity of a city. The book illustrates how roads, whether iconic boulevards or historic highways, become integral elements that define the cultural and aesthetic aspects of urban spaces. This highlights the importance of considering the visual and symbolic impact of roads in urban planning.

Economic Considerations-The book likely discusses how roads influence economic activities within cities. Effective road infrastructure can enhance accessibility, promote commerce, and impact property values. On the other hand, congestion and poorly planned roads may pose economic challenges. The implications for local businesses, property markets, and overall economic vitality are likely addressed in the context of urban development.

Planning Strategies-The book provide insights into planning strategies for cities to navigate the challenges associated with road infrastructure. This may involve discussions on sustainable urban planning, incorporating alternative transportation modes, and addressing the trade-offs between accommodating cars and fostering more environmentally friendly and pedestrian-friendly urban spaces.

2.4 THEORITICAL FRAMEWORK

A theoretical framework is a conceptual structure that provides a systematic and organized way to formulate, develop, and present a set of interconnected concepts, principles, or variables. It serves as the foundation for understanding and analysing a particular phenomenon or problem within a specific field of study.

a) The Bid Rent Theory

William Alonso developed the theory in 1960 (Alonso, 1960). Land uses are classified as residential use, commercial and industrial use, transportation, educational, public purpose, recreational, public utilities, deferred land and agriculture all. These land-use types compete for space in any urban setup, and therefore, they would compete to be located near transportation networks. However, this may not be the scenario for most land uses since the distance from the road determines the cost of land, thereby determining the type of land use that will afford the space. The land use type which occupies the most accessible site will pay the highest rent. These land uses often generate more income and, therefore, can pay the high rent; hotels, supermarkets, and high-income businesses are competitive in rent payments.

Classical bid rent theory by (Alonso, 1964) highlights the fact that there is a decline in rents outwards from the Central Business District (CBD) to set off the reducing revenue production capacity and higher costs such as cost of moving from one point to another. He attempted to apply accessibility requirements to the city centre for various land uses such as residential, commercial, and industrial land use (Narvaez & Griffiths, 2013). He investigated the bid rent theory by studying the relationship between land price and accessibility based on the proposal that distance is not a measure from place to place, rather, it is shaped by the network of streets. The findings established that remote access attracts industries and residential facilities due to the low land value and less disturbance from other activities. He also presumes that due to the relative cost of open land space in locations closer to the
CBD, there are chances that higher-income households are located in the suburbs (Watkins, 2000).

Alonso further hypothesized that accessibility is an element that can harmonize, disperse and accommodate the generation of movement since distance constructs chances for socio-economic interaction. Therefore, access relates to the quantifiable distance that makes a place with a specific function located with another.

The linear urban pattern often emerges as a result of historical factors, such as the development of transportation routes. Roads, railways, or rivers can create a linear structure as the city grows along these axes. Bid rent theory helps to make sense of this by showing how the economic forces of competition for prime locations shape the spatial distribution of land uses along the linear axis.

As one moves away from the CBD along the linear urban pattern, the bid rent decreases, reflecting diminishing accessibility and desirability. Lower-density residential areas and possibly industrial zones emerge in these outer regions. This transition in land use is a manifestation of the economic principles outlined by bid rent theory. It highlights how the spatial organization of a city adapts to the varying economic value of land based on its proximity to the CBD.

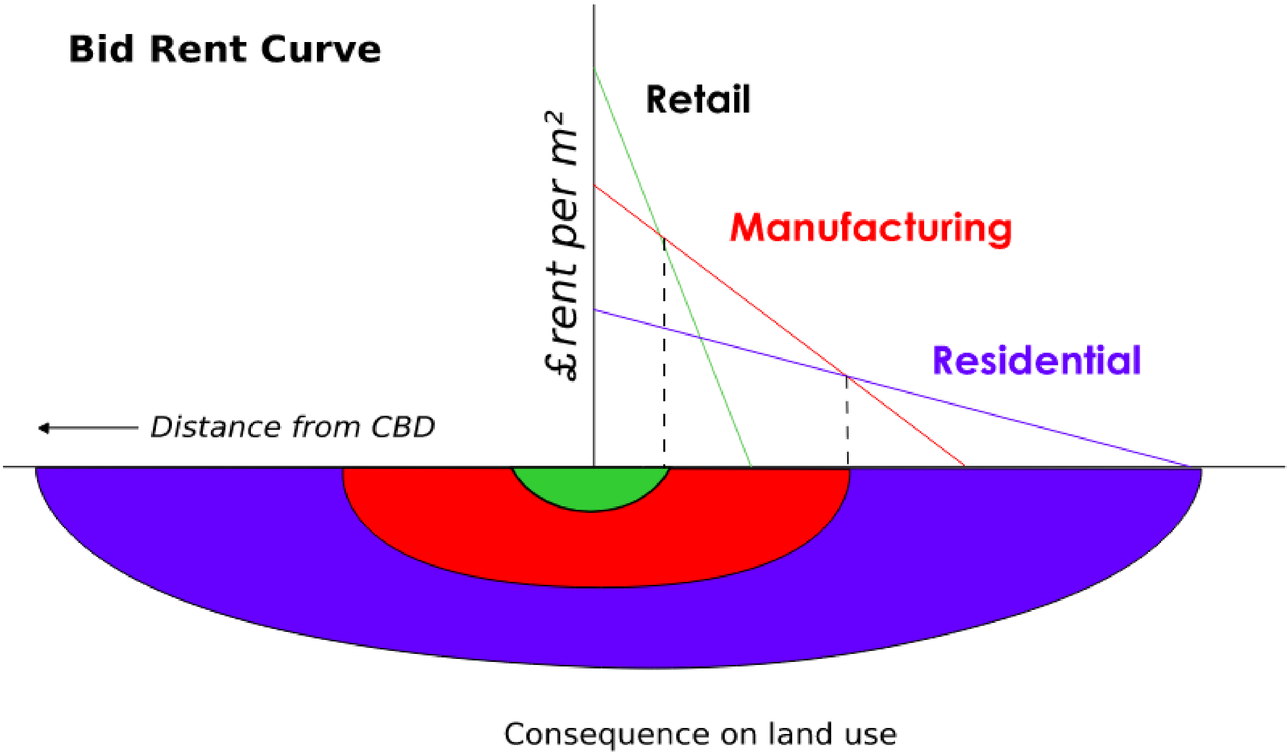

Figure 0:1 Alonso (1964) bid-rent model

The linear urban pattern, when viewed through the lens of bid rent theory, underscores the importance of understanding the economic rationale behind the development of cities. The theory provides a framework for comprehending why certain activities cluster in specific locations and why the intensity of land use changes along the linear axis. In essence, bid rent theory offers insights into the dynamic interplay between economic forces and spatial organization in cities with a linear layout.

Moreover, bid rent theory can be instrumental in urban planning by informing decisions related to zoning, infrastructure development, and land use policies. Recognizing the economic drivers that underlie the spatial distribution of activities can guide planners in creating more efficient and sustainable urban environments.

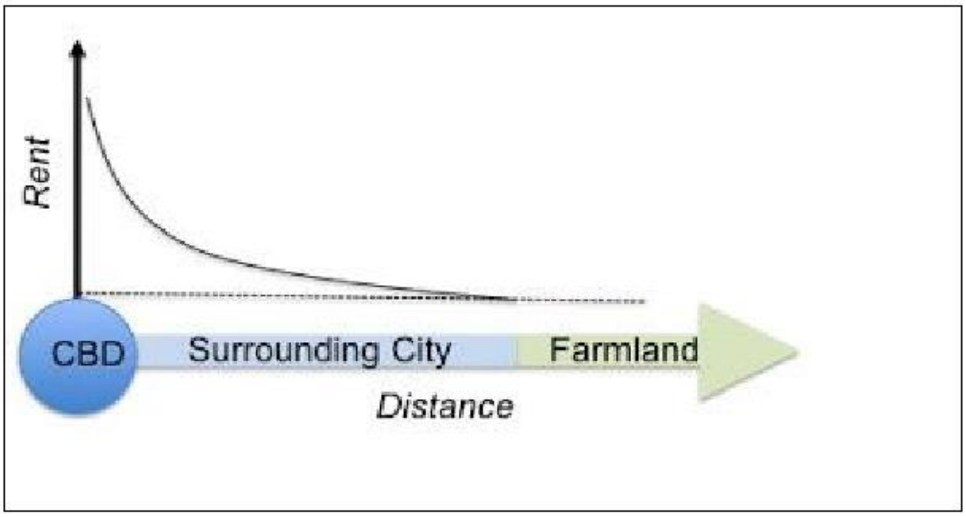

Figure 0:2 Relationships between Rent and Distance Source: Friedrichs, 2012

b) Linear City Model

The Linear City Model, proposed by economist Harold Hotelling in 1929, is a theoretical concept that envisions urban development occurring along a single, linear axis rather than in a traditional circular or concentric pattern. In this model, businesses and residents are situated along a straight line, with the central idea being the optimization of economic advantages, especially in terms of transportation and accessibility.

Key features of the Linear City Model include:

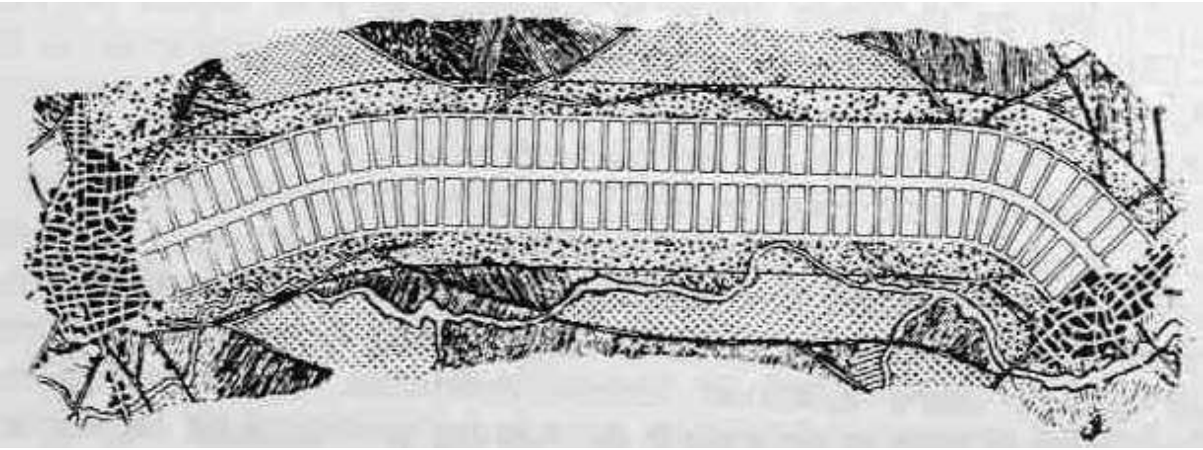

Figure 0:3 The Arturo Soria first plan for a project in neighbourhood Madrid

Transportation Axis

The model assumes the presence of a central transportation route, such as a road or railway, along which development is concentrated. This transportation axis is crucial for facilitating movement and trade, making it a focal point for businesses and services.

Optimization of Accessibility

Businesses and services are strategically located along the linear axis to maximize accessibility and visibility. This arrangement allows for efficient transportation of goods and people along the route, contributing to economic efficiency.

Economic Rationality

The placement of businesses and residents is driven by economic rationality. Entities aim to position themselves in locations that offer the best economic advantages, considering factors such as proximity to markets, competitors, and transportation hubs.

Spatial Concentration

Development is concentrated along the linear axis, creating a distinctive spatial pattern. This concentration is based on the idea that businesses and services benefit from being close to each other and the primary transportation route.

Linear Nature of Development

Unlike other urban models that envision radial or concentric expansion, the Linear City Model suggests that development primarily occurs along a single line. This linear nature simplifies transportation logistics and creates a straightforward spatial structure.

The Linear City Model is often applied to situations where a prominent transportation route, like a major road or railway, influences the layout and growth of urban areas. It provides insights into how economic considerations and accessibility factors can shape the spatial organization of cities, making it relevant for understanding patterns of development along linear corridors such as Magadi Road in Ongata Rongai.

2.5 POLICY AND LEGAL FRAMEWORK

2.5.1 Sustainable Development Goal 2015

The Sustainable Development Goals (SDGs) of 2015, particularly Goal 11, which focuses on making cities inclusive, safe, resilient, and sustainable, recognizes the significant impact of road infrastructure on urban spatial patterns. SDG 11 stresses the need for accessible and sustainable transportation systems, including road networks, to enhance the connectivity and efficiency of cities. The goal encourages the development of urban areas that are well-planned, with an emphasis on sustainable and inclusive infrastructure to reduce inequalities, promote economic growth, and ensure environmental sustainability.

It highlights the role of sustainable urban planning in shaping spatial patterns. It calls for cities to adopt integrated policies and approaches that consider the impact of road infrastructure on land use, accessibility, and the overall structure of urban areas. The SDGs emphasize the importance of creating urban environments that are not only well-connected through roads but also prioritize the well-being of residents, social equity, and environmental conservation. In essence, the SDGs advocate for a holistic approach to road infrastructure development, recognizing its profound influence on the spatial patterns of cities and the achievement of sustainable urban development.

2.5.2 African Agenda 2063

The African Agenda 2063 just appeal that road infrastructure would play a crucial role in shaping the spatial patterns of urban areas. Improved transportation networks, including roads, are vital for enhancing connectivity, facilitating economic activities, and promoting sustainable urbanization.

2.5.3 Kenya Vision 2030

The 2030 vision aspires for a country firmly interconnected through a network of roads, railways, ports, airports, water and sanitations facilities and telecommunication. Furthermore to ensure that the main projects under the economic pillar are implemented, investment in the nation's infrastructure is given the highest priority which in hand attract development.

2.5.4 Kenya National Spatial Plan

One of the key principles of National Spatial Plan is to promote public transportation: this will favour public transportation over private transport to ensure efficiency and functionality of urban places. This will ensure that developments are attracted in those areas.

2.5.5 Kajiado County Integrated Spatial Plan

Integrated strategic urban development planning is about future-oriented development that gives equal consideration to economic, environmental, social and cultural dimensions of the sustainable city.

2.5.6 Physical and Land Use Planning Act, 2019

The Act was established to help with land development matters by giving guidance on planning, usage and regulation of land. The objects of the Act are to provide: procedures and levels for the preparation and implementation of physical and land use development plans at the national and county level, guidance and oversight of physical and land use planning, strategies for development control and the policing of physical planning and land use, a strategy for the coordination of physical and land use planning by county governments, a strategy for fair and viable use, planning and management of land and a framework to ensure that investments in property benefit local communities and their economies(Kenya Law Reports, The Physical and Land Use Planning Act, 2019, 2019).

2.5.7 Urban Areas and Cities Act 2011 CAP275

The Act was established as an Act of Parliament for the purpose of categorization, administration and the running of urban areas and cities, to give basis of setting up of urban areas, to contribute to the concept of administration and management of urban areas and involvement of residents in the administration and management of cities.

The Urban Areas and Cities Act places the running of cities and municipalities in the County Government. This is to be administered through a Board that has been enacted as per the Act. The board performs other functions such as formulating and implementing plans and, in the process, ensuring the involvement of the users. Section 36 of the Act emphasizes on incorporation of all functions through the integration of city development plan which should be binding, guiding and informing all planning and development decisions. Furthermore, Section 37 of the Act lays down guidance on how an urban area integrated development plan should be in line with the county government's development plans and master plan (Kenya Law Reports, Urban Areas and Cities Act No 13 of 2011, 2011) advocates that cities and municipalities set under the Act are to work within a merged development planning framework that will be the premise for development control and provision of services such as water, health, electricity and solid waste management.

This Act therefore outlines the overall oversight role that is to guide the urban development and planning of land uses which also need to be in line with the Kenya Roads Act (No 2 of 2007).

2.5.8 County Governments Act 2012

Section 104 of the County Governments Act 2012 requires a county government to plan for the county and ensures that public funds are used appropriately within the planning structure that incorporates physical, economic, social, environmental and spatial planning. The law also indicates that the county government selects county departments, sub-counties and ward representatives as people in charge of planning and that county plans to be obligatory to all sub-counties.

The law indicates that city plans are the instruments for development control within the city and that uses and basis of land use and building plans and placement of infrastructure are to be provided for by the plans. Monitoring and compliance with the plan are an important aspect of execution which the county government has to adhere to and therefore, create land use planning unit to undertake it.

2.5.9 National Land Commission Act No 5 of 2012

The commission has been mandated as in Article 67 of the Constitution of Kenya, to keep track of and supervise plan implementation in cities in order to cushion the sovereignty of the users and promote federalism hence, they obtain continuous reports from the county government board and carry out visits to confirm the conditions. The National Land Commission also seeks to promote cities which are intentionally aggressive, habitable, usable, economically vibrant, socially inclusive and environmentally friendly. This is to be achieved through integrated and functional urban transport.

The Sessional Paper Number 3 of 2009 on National Land Policy recommends that: in order to prepare land use plans at different levels of the government, there is need to adhere to examination of aspects such as the economy, safety, aesthetics, and harmony in land use and environmental sustainability.

From the above laws and legislation, it is clear that counties have been given the mandate of developing various land uses within their regions but this should be in line with the national policies and regulations on land use. Therefore, there is need to align the national road development plans with county land use development plans.

2.5.10 Kenya Roads Act, 2007

It provides a comprehensive legal framework for the development, management, and maintenance of roads in Kenya. It establishes the Kenya National Highways Authority (KeNHA) and the Kenya Rural Roads Authority (KeRRA) as key institutions responsible for the planning and execution of road infrastructure projects. The Act outlines the powers and functions of these authorities, the procedures for road construction and maintenance, and mechanisms for funding road projects. It also addresses issues related to the regulation of traffic on roads and sets standards for road design and construction. The overarching goal is to ensure a well-maintained and efficient road network to support economic development and enhance transportation across the country.

The legal framework that guide land use development in urban areas

2.5.11 Transport-oriented development planning (TOD)

TOD is a planning and design strategy that encompasses land use and transport planning and seeks to include walkable and liveable neighbourhoods with high density of land uses that are mixed so as to achieve sustainable urban growth. It aims at advancing compact, mixed-use, pedestrian friendly urban development that closely intertwines with mass transit by clustering jobs, services and amenities around public transport areas (Cervero & et al, 2004).

Consolidating transport and land use planning contributes to the sustainability of cities and promotes the use of public transport. It is viewed by many as a promising tool for curbing congestion and the automobile dependence it generates and hence has a particular relevance to integrated development of public transport and urban pattern. This will allow for goods and services to be readily available and accessible to neighbourhoods and therefore minimize the distance of travel to work. This will reduce the impact of traffic snarl ups to the core town and will also promote infrastructure development within neighbourhoods. TOD brings many benefits including ease of access to transit, making it easy to move around without a car

hence this discourages traffic congestion in urban areas. This strategy allows for better planning of neighbourhoods by incorporating all land uses in one area.
Development should be coordinated in advance to avoid inconsistent patterns as it will be difficult to convert land use once land is built-up, hence planning is necessary to harmonize public and private investment decisions. On a smaller scale, this could include the construction of low density development around a road network, which at later point prevents densification. On a larger scale, this could entail offices being built far away from public transport nodes therefore leaving the public transport system operating below capacity while creating congestion on roads.

2.6 CONCEPTUAL FRAMEWORK

It involves identifying and defining the key concepts such as "road infrastructure," "urban spatial pattern," and any other relevant factors. It helps to establish how these concepts are related and how they contribute to the research objectives.

1. **Investigating the Role of Transport Infrastructure in Creating the Linear Urban Pattern along Magadi Road**:

This posits that transportation infrastructure shapes urban development patterns by influencing land use decisions and spatial organization. It suggests that the location and design of transportation networks, such as roads and transit systems, can generate distinct urban forms, including linear patterns along major transportation corridors like Magadi Road.

Concepts such as accessibility, mobility, and spatial accessibility can be used to analyze how the layout of Magadi Road influences land use patterns, economic activities, and residential development along its route.

2. **Examining the Effects of the Linear Pattern on Provision of Services and Utilities**:

Along Magadi Road, the linear urban pattern may lead to the concentration of commercial establishments and service providers, affecting accessibility to services for residents and businesses.

Analysis using concepts such as market area analysis, service catchment areas, and spatial equity can help assess the distribution of services and utilities along Magadi Road and identify gaps or disparities in access.

3. **Examining the Socioeconomic Outcomes of the Linear Pattern**:

Urban economics focus on understanding how urban spatial patterns influence economic activities, employment opportunities, and housing markets. Along Magadi Road, the linear pattern may impact property values, employment opportunities, and income levels in adjacent areas.

The relationship between spatial patterns and social processes, including segregation, gentrification, and community dynamics. Analysis can explore how the linear pattern affects social interactions, cultural diversity, and neighbourhood cohesion along Magadi Road.

4. **Providing Mitigation Strategies to Address the Challenges Arising from the Linear Pattern**:

Sustainable urban development advocate for strategies to mitigate negative impacts of urbanization, promote environmental sustainability, and enhance quality of life. Along Magadi Road, mitigation strategies may include promoting mixed-use development, improving pedestrian and cycling infrastructure, opening of corridors along magadi road and enhancing public transit options to reduce reliance on private vehicles.

Transportation planning emphasize the importance of integrated, multimodal transportation systems that prioritize efficiency, safety, and accessibility. Mitigation strategies can leverage concepts such as transit-oriented development, traffic calming measures, and land use planning to address congestion, pollution, and inequities associated with the linear pattern along Magadi Road.

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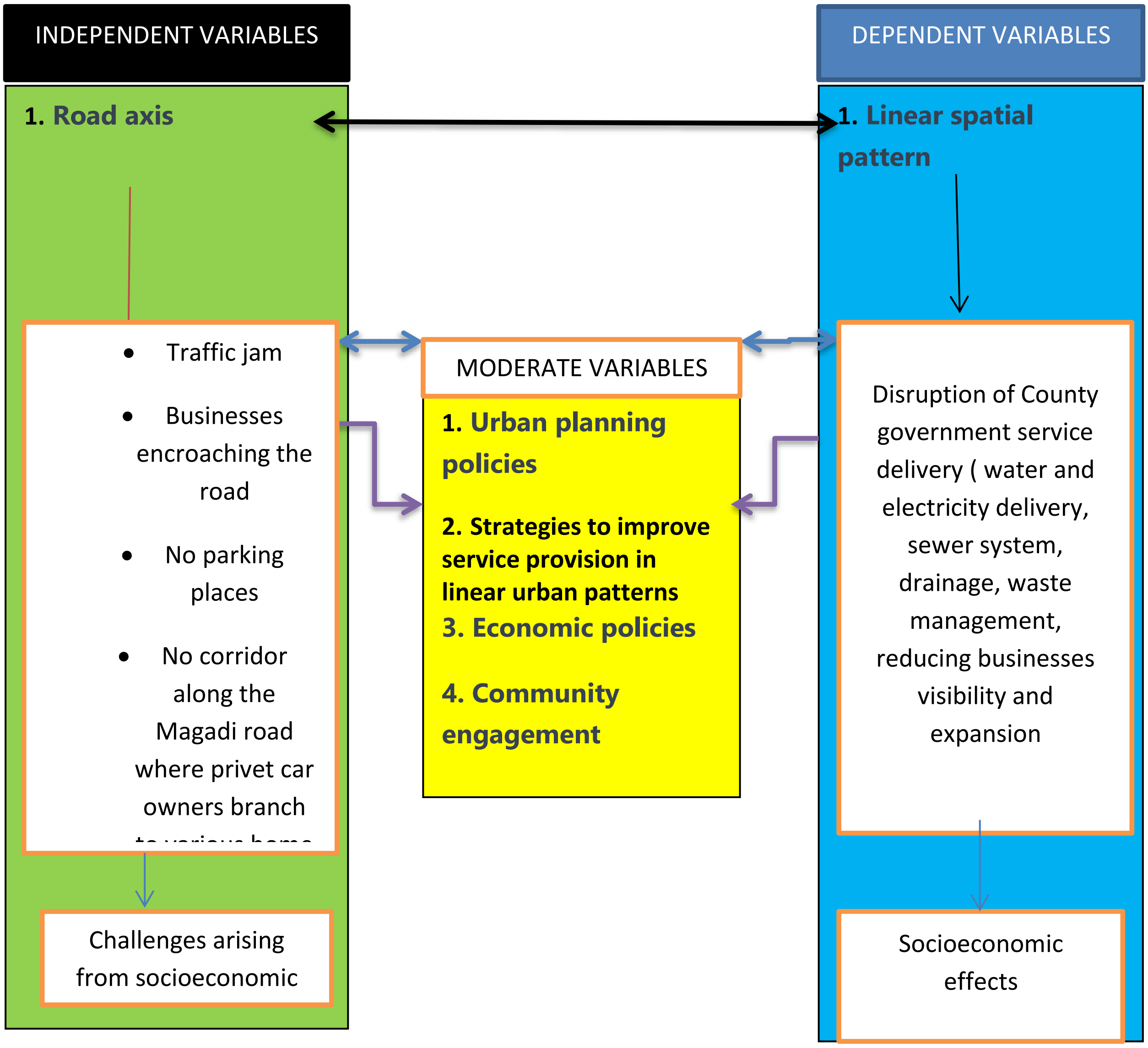
Conceptual framework figure



INDEPENDENT VARIABLES
DEPENDENT VARIABLES
1. Road axis
1. Linear spatial pattern
Traffic jam
Businesses encroaching the road
No parking places
No corridor along the Magadi road where privet car owners branch
MODERATE VARIABLES
1. Urban planning policies
2. Strategies to improve service provision in linear urban patterns
3. Economic policies
4. Community engagement
Disruption of County government service delivery (water and electricity delivery, sewer system, drainage, waste management, reducing businesses visibility and expansion
Challenges arising from socioeconomic
Socioeconomic effects

CHAPTER THREE: METHODOLOGY

3.0 Introduction

The chapter deals research methodology, which will be used to carry out the study. This includes research design, population, samples and sampling procedures, instruments, data collection and data analysis procedures.

3.1Research design

The research design used is mixed-method design. The main purpose of this method is to find relation between the independent variables (transport infrastructure) with the dependent variables (urban spatial pattern. The approach incorporates both qualitative and quantitative methods. Quantitative survey dealt with the rate of growth and the current population of the study area while qualitative dealt with other factors for the linear urban growth pattern, challenges and opportunities which is collected using questionnaires administered to the study population and key informants which is done in the department of physical planning offices of Kajiado County.

3.2 Target Population

It is calculated by using the Slovin's method using the total population of household Ongata Rongai town 2019 statistics on which it had 101,378 households on which the formula below is;

n=N

$1+N(e)^2$

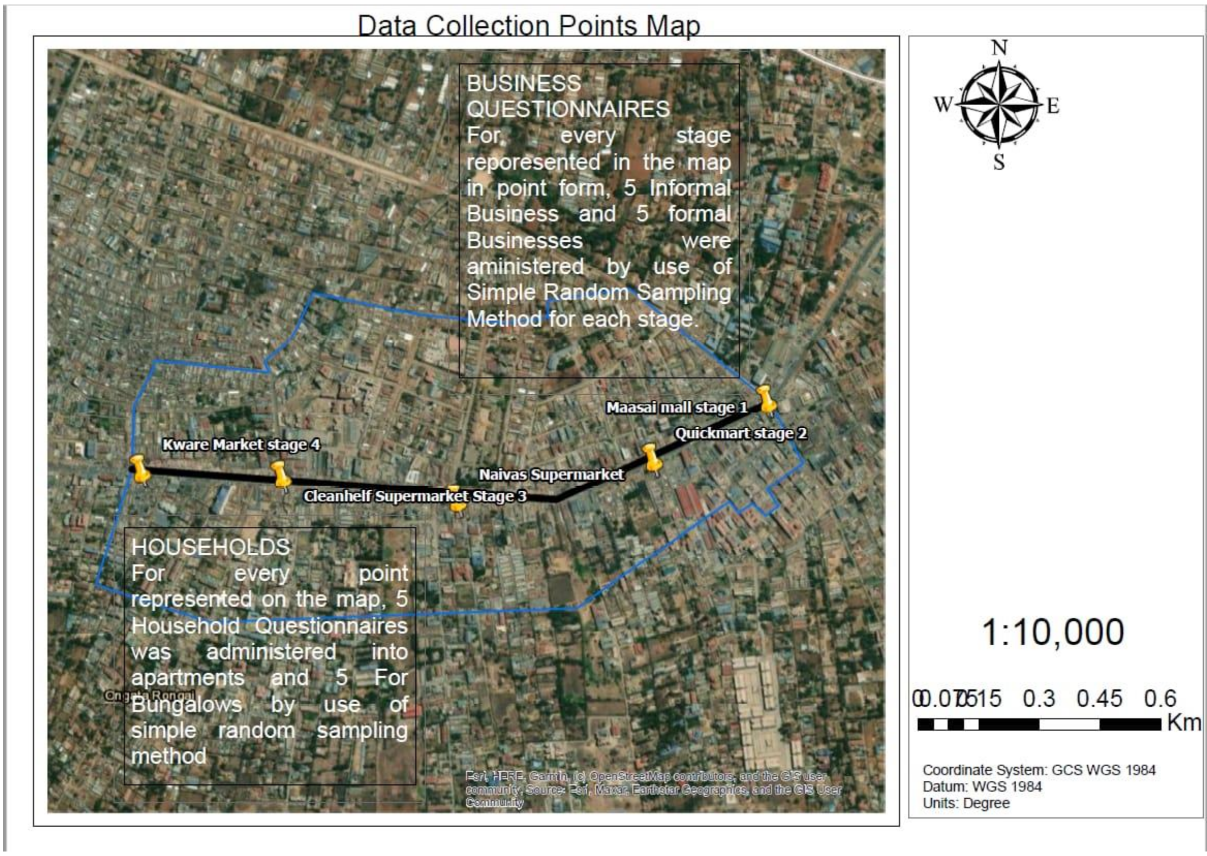
n=Sampling size

N=number of Households

e=margin of error which is 10%

According to the formula, I was supposed to administer questionnaire to 1,014 houaeholds but given that am using along Magadi road from Maasai Mall to Kware market plus due to lack of resource and limited time, I used 100 people which were households and business owners along the Magadi road as my target sample size by 50/50. These involved administering household in every stage and businesses along the same stage by use of purposive methods.

The map used is as shown below



3.3. Data collection method

Methods of collecting data were classified in two ways as follows:

3.3.1 Primary Data

The primary data was collected through random sampling of business people along Magadi road and households. Moreover, a Key Informant Interview was conducted to urban management person and transport personnel in the Physical Planning sector of Kajiado County. Other data collected is for verification and ground truth by observation since the area being studied is large, simple random sampling was done for every class represented for land use and land cover change. This was done by used of questionnaires and observation matrix. Photographs too were also taken to help in observing the current state of Magadi Road, Linear pattern and various land use and land cover activities which are current on the ground.

3.3.2Secondary Data

A desktop review was conducted to conceptualize the concept linear urban pattern, undesirable growth of towns how the transport infrastructure has impacted their causes and how they are trying to control their impacts, given that Ongata Rongai growth is highly being influence by is nearest to Nairobi City. Written documents such as journal articles, books, reports, and other documents acquired from electronic sources and the physical location were reviewed. Satellite images were considered significant data because they were used to generate Land Use and Land Cover maps.

Topographic maps were collected to generate basic layers, such as boundaries and roads, as well as to enhance satellite images and for geo referencing purposes. Furthermore, socioeconomic data, such as population data was collected and used.

3.4. Data acquired and source

The area of interest was selected based on the demarcation of Independent Election and Boundary Commission wards (IEBC). The data on country demarcation is available on the IEBC website. Consequently, Google Earth software, an open-source program that allows users to view the earth through high-resolution satellite images, was used to confirm and observe the characteristics of the study area. The data acquired was social, economic and environmental. These were collected using primary sources where questionnaires were administered to sample business people of Ongata Rongai and resource persons. The data was analysing using Kobo collect tool.

The spatial data acquired was geographical maps presenting the location features and boundaries of Ongata Rongai and the surrounding area using Geographical Information System (GIS) and Google Earth images. Population data of Ongata Rongai town was acquired from Census data by the Kenya National Bureau of Statistics (KNBS).

3.4.1Simple random sampling

A baseline survey was conducted to determine factors for the linear growth pattern of Ongata Rongai and the impacts both challenges and opportunities they bring to the road infrastructure and to the surrounding communities. Forms were created using Kobo Collect App and I administered them personally to the various group of people (businesses) along Magadi road. This method enabled me to strategically manage my target audience and they answered the questions at their convenience.

3.4.2Purposive sampling

Key informant Interview was conducted through oral interview. The interview targeted the person who was very resourceful in Physical planning and management of Ongata Rongai. This was to help me identify some of the plans in place for handling the challenges facing Ongata Rongai town due to its linear pattern of growth. The person whom the interview was conducted with was the physical planner of Kajiado County.

3.4.3Observation Matrix

These involve observing the current state of road infrastructure and both social and economic activities in Ongata Rongai town which were recorded through photographs.

TABLE 1.1 DATA NEEDS AND SOURCEES MATRIX FOR THE PROJECT

OBJECTIVES	**DATA NEEDS**	TYPES OF DATA	TOOLS/ INSTRUMENTS FOR DATA COLLECTION	HOW TO ANALYSE THE DATA	HOW TO PRESENT	EXPECTED OUTPUT
To investigate the role of transport infrastructure in creating the linear urban pattern along Magadi road.	Definition and characteristics of a linear urban pattern. How transport infrastructure contributes to the formation of this pattern.	Historical background, Secondary data. Situational analysis, Primary data.	Literature review, Observation Questionnaires Photography	Questionnaire analysis and reports evaluation	Graphs Report writing	Outputs will include maps, correlation analyses, and guidelines to enhance the symbiosis between transportation and the linear urban layout.
To examine the effects of the linear pattern on provision of services and utilities	Discussion on how a linear urban pattern affects access to essential services. Analysis of challenges faced by service providers due to the elongated nature of a linear city	Secondary data Primary data.	Literature review Interview Observation Questionnaires Photograph	Questionnaire analysis by use of Kobo collect tool Report evaluation	Writing of report, graphs, charts, maps	Metrics for service accessibility, community impact assessments, and comparative analyses. Visualizations and documentation of challenges and opportunities,
Examining the Socioeconomic Outcomes	Exploration of social implications associated	Primary data	Questionnaires Interviews observatio	Questionnaire analysis by use of	Writing of report, graphs, tables,	Reports on economic disparities, maps

	with a linear urban pattern. Economic consequences for businesses located within or outside the main transportation corridor. increased traffic congestion along major transportation routes		n	Kobo collect tool	maps, charts.	illustrating educational and employment opportunities, and recommendations for social services.
Providing Mitigation Strategies	Mitigation strategies that can be implemented to overcome challenges	Proposed project prescription	Review of project proposals	Analysis of the reviewed project prescription	Writing of report and maps	Recommendation from the report.

4.5Development Trajectory along Magadi Road

Ongata Rongai town, situated in Kenya, has evolved into a bustling urban centre with a distinct character shaped by its residential, commercial, and educational landscape. As a residential hub, the town accommodates a diverse population, ranging from families seeking suburban living to students attending nearby educational institutions. For instance, neighbourhoods like Rimpa Estates and Maasai Lodge Estate offer a range of housing options, from apartments to standalone homes, catering to different preferences and income levels.

In terms of commercial activities, Ongata Rongai boasts a vibrant market scene and a plethora of shops, restaurants, and service providers. The Maasai Market, held on weekends, offers an array of traditional crafts and goods, attracting both locals and tourists. Additionally, the town features supermarkets like Tuskys and Naivas, providing residents with convenient access to groceries and household items. Small businesses thrive along

Magadi Road and Magadi Stage, offering everything from clothing boutiques to hardware stores.

Education plays a significant role in Ongata Rongai's identity, with several institutions contributing to its status as an educational hub. The Multimedia University of Kenya and Advent Hill University College are notable examples, attracting students from across the country. Additionally, the town hosts numerous primary and secondary schools, such as Ongata Academy and Oloosirkon Primary School, catering to the educational needs of local families.

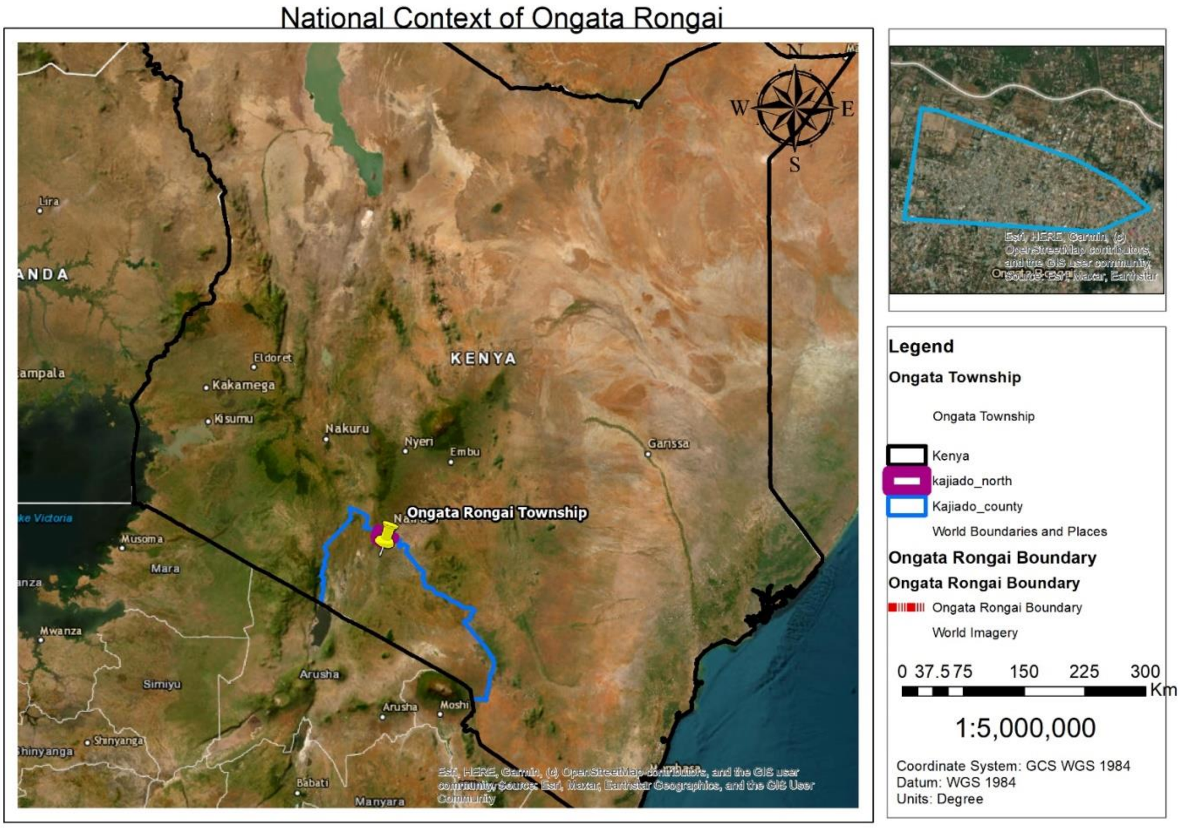
Transportation infrastructure is a key feature of Ongata Rongai, with the town serving as a vital transit point. Matatus (minibuses) ply routes between Ongata Rongai and Nairobi's central business district, facilitating daily commutes for residents and commuters. The Ngong SGR Station, part of the Standard Gauge Railway project, provides further connectivity, offering residents an alternative mode of transportation to Nairobi and beyond.

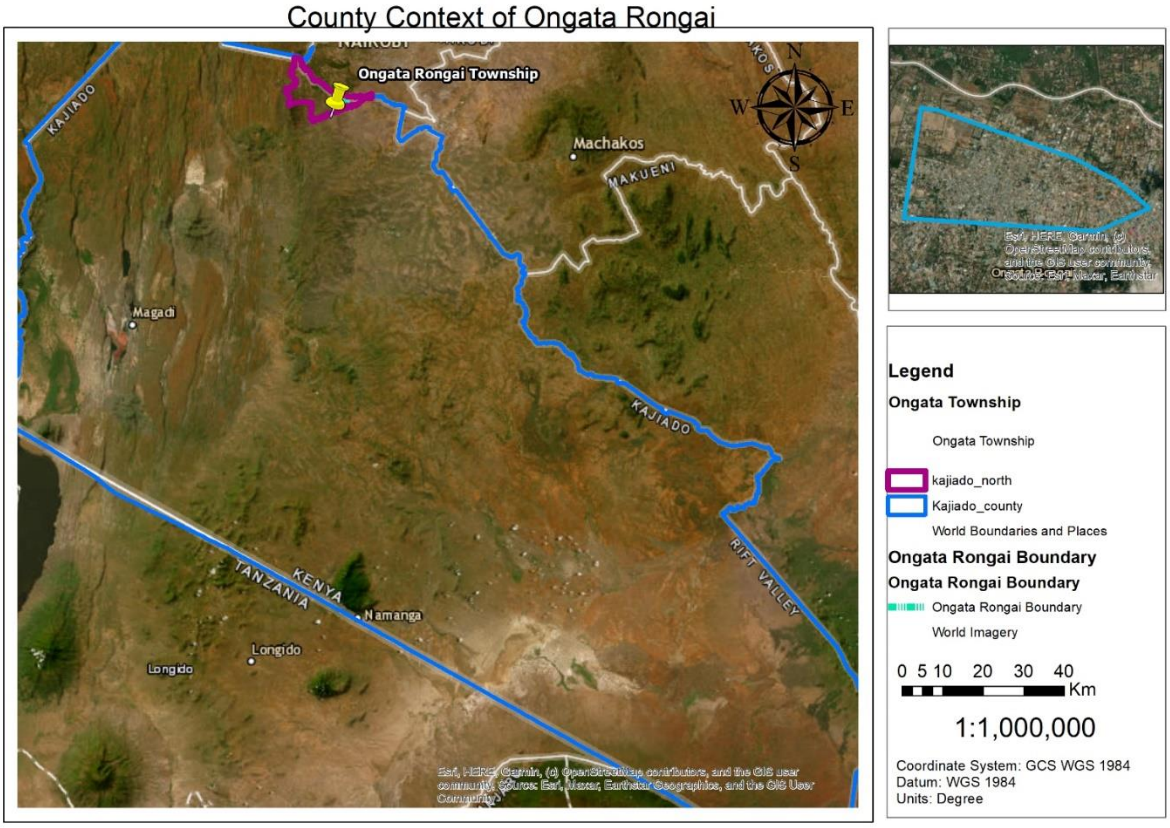
As Ongata Rongai continues to grow, ongoing developments in infrastructure and services aim to enhance the quality of life for its residents. Projects such as road expansions, water supply improvements, and recreational facility upgrades underscore the town's commitment to sustainable urban development. Overall, Ongata Rongai stands as a dynamic urban centre, blending residential tranquillity with commercial vibrancy and educational opportunities, making it a significant contributor to the socio-economic landscape of the region.

CHAPTER FOUR: THE STUDY AREA

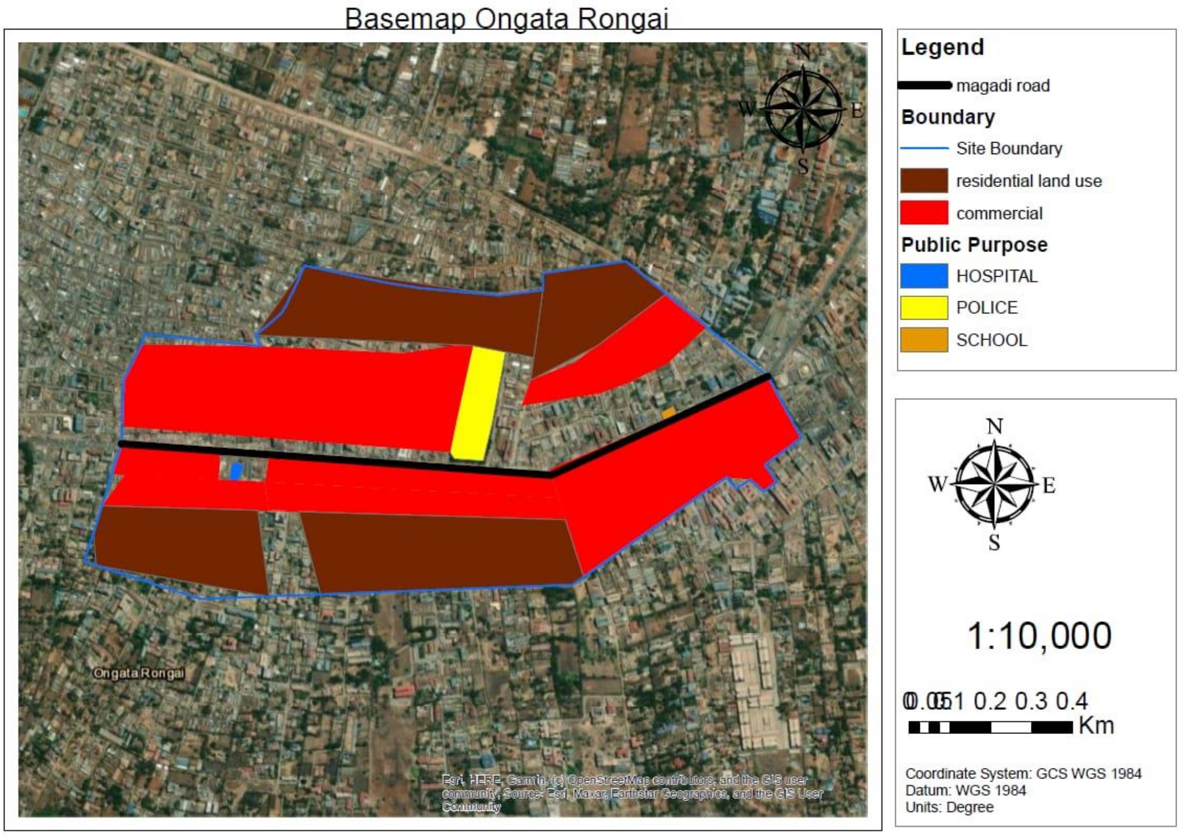
Ongata Rongai Town, Kajiado County

The study area is located in Ngong Municipality, Kajiado County, 17km south of Nairobi the capital city, and west of Ngong. Ongata Rongai town is located at the following coordinates: 1.3939^{0}S and 36.7442^{0}Eand lies 1731 meters above sea level. It is the most populous town in Kajiado County and the eleventh largest urban centre in Kenya with a population of 172,569 (KNBS, 2019). This is one of the fastest-growing towns in Kenya. The town is also part of the larger Nairobi Metropolitan Region where it serves as the dormitory town for Nairobi City.



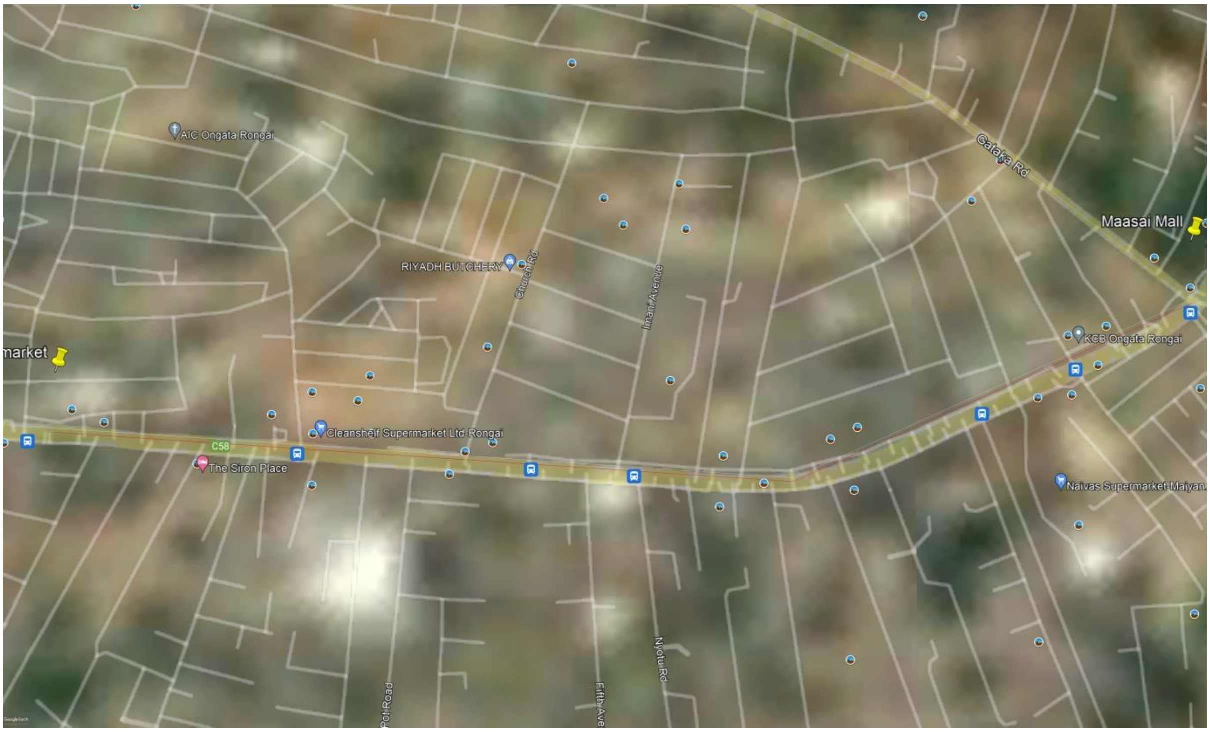


Land use map of the study area from kware market to Maasai Mall



Magadi Road from Maasai Mall to Kware Market Google Earth Imagery situation analysis

1985 situation



2009 situation

2014 situation of development along the road

2019 situation of development

Current situation of development

Historical Background of Ongata Rongai

During Kenya precolonial period, the land was occupied by the Maasai community who were practicing pastoralism. The town developed from a cattle market, on the lower section, and a quarry site, on the upper side of the town and since 1950 the town has been rapidly growing to become the current township. Furthermore, due to Rongai's proximity to the capital city, intense development has been taking place since the 1990s due to the huge demand for housing by people working in the city. This has also resulted in the sprawling of the settlement beyond the border of Rongai town affecting the rural community who were formally nomadic pastoralists and forcing them to settle in addition to having alternative ways of managing cattle on small parcels thereby increasing land degradation. This has also reduced flora and fauna and increased human-wildlife conflict(County Government of Kajiado, 2019; Emily, 2000) The town is currently occupied by a population that is multi-ethnic due to immigration that has been taking place.

Through time the land tenure system has changed from community land and group ranches to individual ownership (Morara et al., 2014). There has been increased crop production and intensified livestock farming by the immigrants while around the peri-urban area of Rongai town, there has been an increased variety of land uses. The change inland use land cover has been due to intensified population pressure coupled with a lack of proper management of the town.

UN-HABITAT et al., 2020 report that Ongata Rongai is experiencing a high-density population due to its proximity to the capital city growing at a rate of 5.5% p.a. Kazungu et al., 2011 study indicated the urban management challenges faced in Ongata Rongai and these are: One, the failure of the legal and institutional framework to support information sharing and partnership between intergovernmental agencies in Kajiado County.

Two is that Ongata Rongai is faced with inadequate human and fiscal resources, and there are also bureaucratic planning systems that make planning structures and processes in both areas incapable of dealing with the scale of planning and infrastructure problems confronting them. Three is the challenge of a dependency syndrome of the donor precipitating the adoption of unsustainable solutions. Such that developments only take place when there is support from a donor. Four is the growth patterns are uncontrolled with informal commercial activities undertaken on road reserves and junctions which is the one of key aspect am assessing in this research. These challenges have occurred due to the administrative and institutional inability to plan and control the rapid growth of development. Finally, there is limited public participation of the community in the planning and management of Ongata Rongai.

CHAPTER FIVE: RESULTS, FINDINGS AND DISCUSSIONS

5.0 Introduction

In this chapter, the findings from the analysis of questionnaire data are presented. The analysis of the data was guided by the objectives of the research. Each variable's data was evaluated using descriptive and inferential statistics, with the results summarized in tables and figures, and their significance explained.

5.1 Questionnaire Return Rate

The research was conducted on sample of 100 respondents from the target group that is, 59 questionnaires were administered to households and 50 were administered randomly to businesses along Magadi road from Maasai Mall to Kware market within the area enclosed in the spatial frame of the study. The statistics analysed were used to show the relationships between variables without any bias towards any target group.

Response Rate table

No of questionnaires	Return for businesses	Return for households	Total Return no	Response rate
100	45	41	86	86%

The response rate was 86 percent, with 86 out of 100 surveys being completed as shown in the table above. This level of participation is adequate for drawing research findings.
There was a high response rate to the questionnaire (86 percent) since the researcher in person conducted the administration of the instrument.
According to Mugenda & Mugenda, this is acceptable (2003). This strategy not only decreased the impacts of language barrier, but also secured a high instrument response and scoring rate by ensuring that respondents' questions about clarity were addressed at the time of data collection.

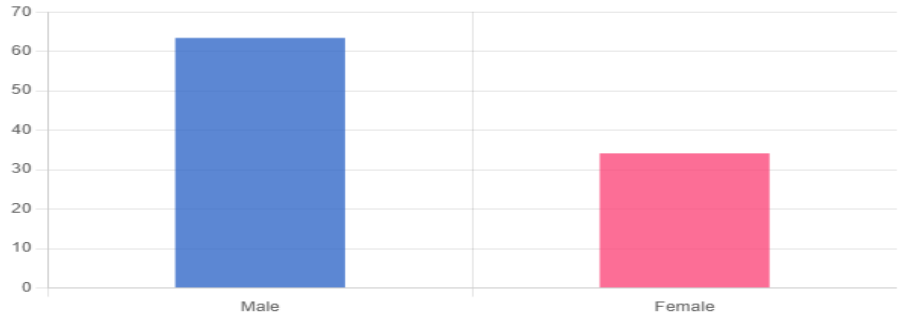
5.2 Socioeconomic Profile of Respondents

This section presents the key characteristics of the respondents in this in terms of gender of the respondent, age bracket, employment status, areas of work, the period they have lived in this area and average monthly income.

5.2.1 Gender

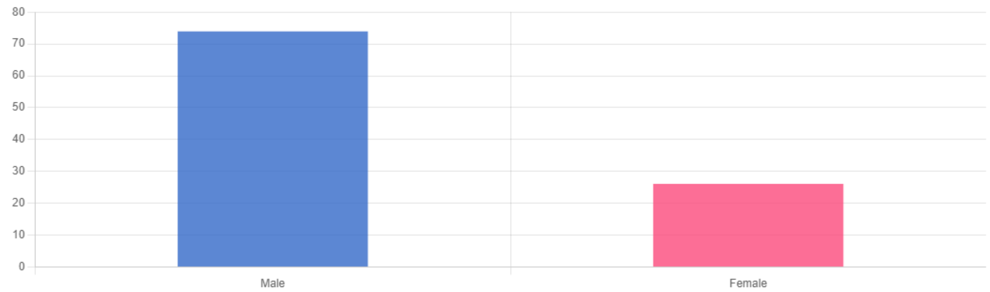
The study sought to establish the gender of the respondents' parents/guardians. This is important because the conclusion reached with one gender group might not be representative of experience of the other group. The findings are as shown in the figure below both for households and businesses.

For household questionnaires



Value	Frequency	Percentage
Male	26	63.41
Female	14	34.15

For business questionnaires



Value	Frequency	Percentage
Male	34	73.91
Female	12	26.09

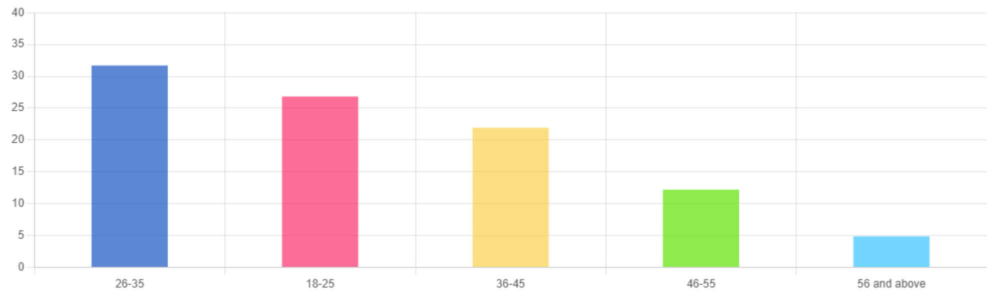
From the study findings, most (67.7%) of the respondent were males, while 30.2% were females. This implies that majority of the respondents who use Magadi Road in Ongata Rongai were males.

5.2.2 Age bracket

The study sought to find out the age of the respondents. There is a strong correlation between age and opinions and/or behaviours and this ensures that the data obtained is not biased towards a particular age group. The findings are as shown in below.

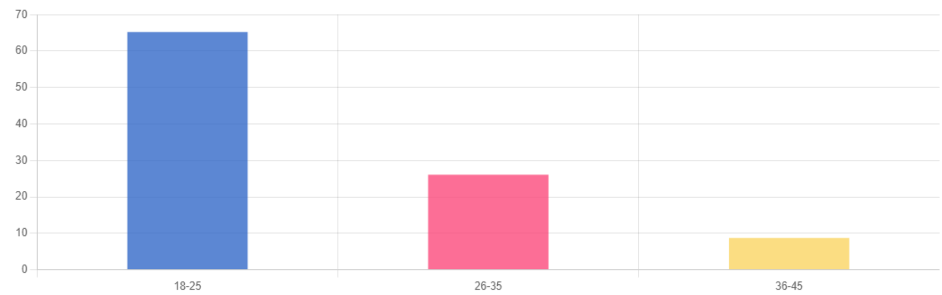
Age Distribution of Respondents

For households questionnaire



Value	Frequency	Percentage
26-35	13	31.71
18-25	11	26.83
36-45	9	21.95
46-55	5	12.2
56 and above	2	4.88

For business questionnaires



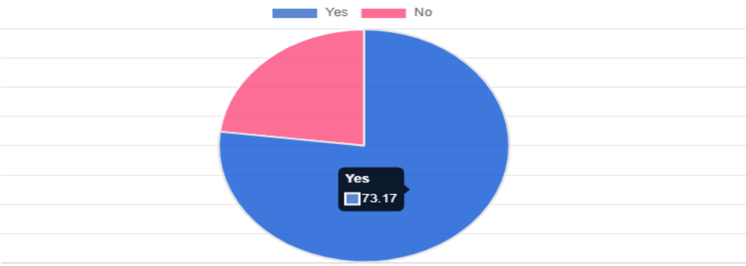
Value	Frequency	Percentage
18-25	30	65.22
26-35	12	26.09
36-45	4	8.7

Based on the data collected, 47.6% of respondents were between the ages of 18 and 25, 29% were between the ages of 26 and 35, 5.8% were between 46 and 55, 4.6% were between the ages of 36 and 45, 2% were between the ages of 56 and above.
This suggested that the majority of respondents were between the ages of 18 and 25 in that order as written above.

5.2.3 Employment status for the resident

Understanding the employment status of residents in this research project is crucial for demographic profiling, policy implications, economic analysis, and targeting social welfare services effectively. It also helps in shaping the research scope and focus to address pertinent issues related to employment dynamics and their impact on society. This was address to household questionnaires and below was the results.

Employment status with yes representing employed while no represent not employed.



The pie chart indicates that 73.17% of the residents which I administered were employed with only 26.83% were an unemployed.

5.2..4 Place of work

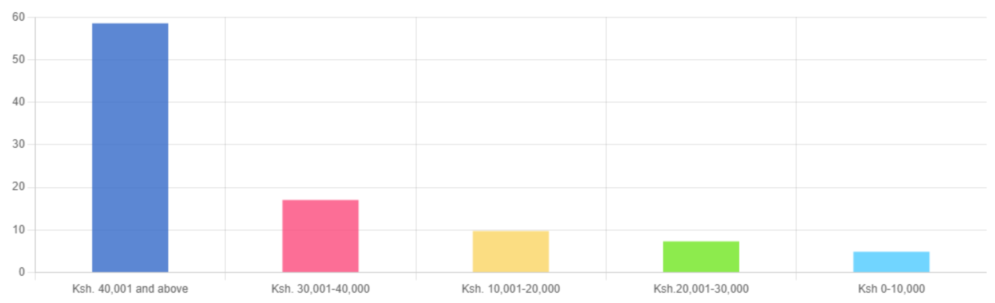
Knowing the place of work for residents of Rongai, in this research project is essential for understanding commuting patterns, local economic dynamics, assessing transportation infrastructure needs, and tailoring employment-related interventions to address specific challenges within the community effectively. The finding shows that most people commute to go to work either in Nairobi City or neighbouring county.

The table shows places of work for resident of Ongata Rongai

Value	Frequency	Percentage
Nairobi	15	36.59
Rongai	10	24.39
Machakos	4	9.76
Makueni	2	4.88

5.2.5 Average monthly income

Understanding the average monthly income for residents of Rongai, in this research project is crucial for assessing the standard of living, economic inequality, purchasing power, and informing policy decisions related to income support, social welfare programs, and economic development initiatives tailored to the needs of the community. The results show that most of the residents I administered are leaving above poverty line which in Kenya is around Ksh. 9,000 per month.



Value	Frequency	Percentage
Ksh. 40,001 and above	24	58.54
Ksh. 30,001-40,000	7	17.07
Ksh. 10,001-20,000	4	9.76
Ksh.20,001-30,000	3	7.32
Ksh 0-10,000	2	4.88

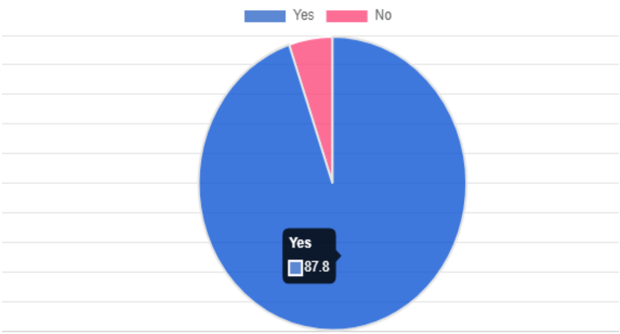
The findings also indicated that most people administered had stay in that area for more than 10 years experiencing and witnessing the changes in development that had been occurring over time.

5.3 Findings according to the four objectives

This outlines the outcomes asper the questionnaires analysis, observation and comments as per the key informant.

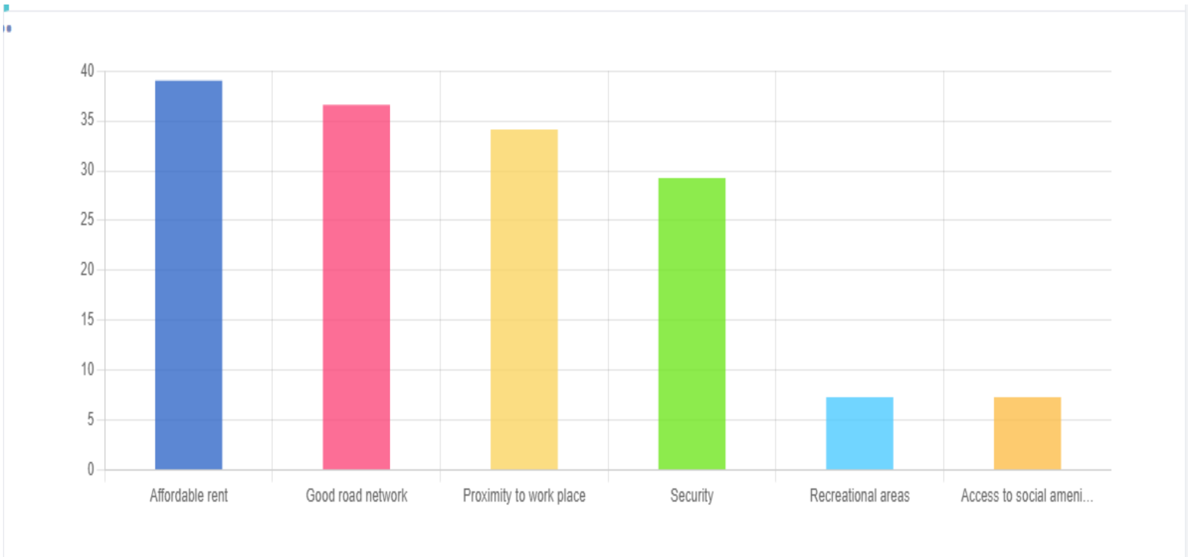
5.3.1 To investigate the role of transport infrastructure in creating the linear urban pattern along Magadi road.

The first and very important thing was to find out whether the residents were able to observe the linear growth of Rongai along Magadi road. The findings indicated that over 87% could observe such kind of the phenomenon growth as shown be



Some of the key roles of transport infrastructure in creating the linear urban pattern along Magadi Road in Ongata Rongai as from the survey are as follows;

Factors that attracted settlement along Magadi Road as per the survey

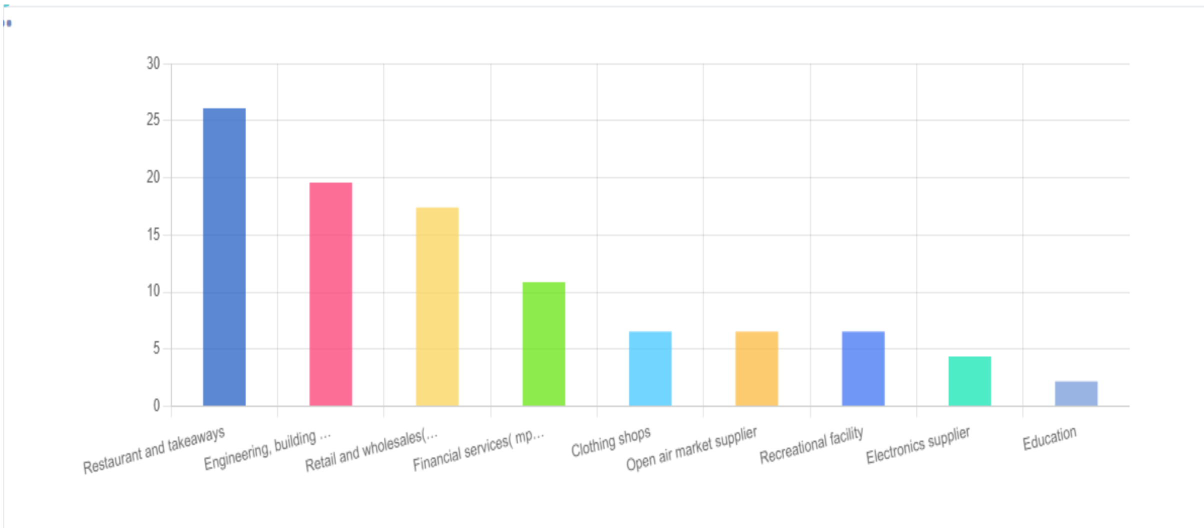


Accessibility and Connectivity

The presence of Magadi Road as a major transport route has fundamentally transformed Ongata Rongai's urban landscape by providing essential accessibility and connectivity. Serving as a vital link between the area and the neighbouring regions, including Nairobi city centre, Machakos, Makueni, Magadi Road acts as a channel for people and goods. This accessibility attracts businesses, residents, and services to establish themselves along its length, forming the foundation of the linear urban pattern observed today. Without this crucial transport artery, the development of Ongata Rongai as a thriving urban centre would have been severely limited.

Commercial Development

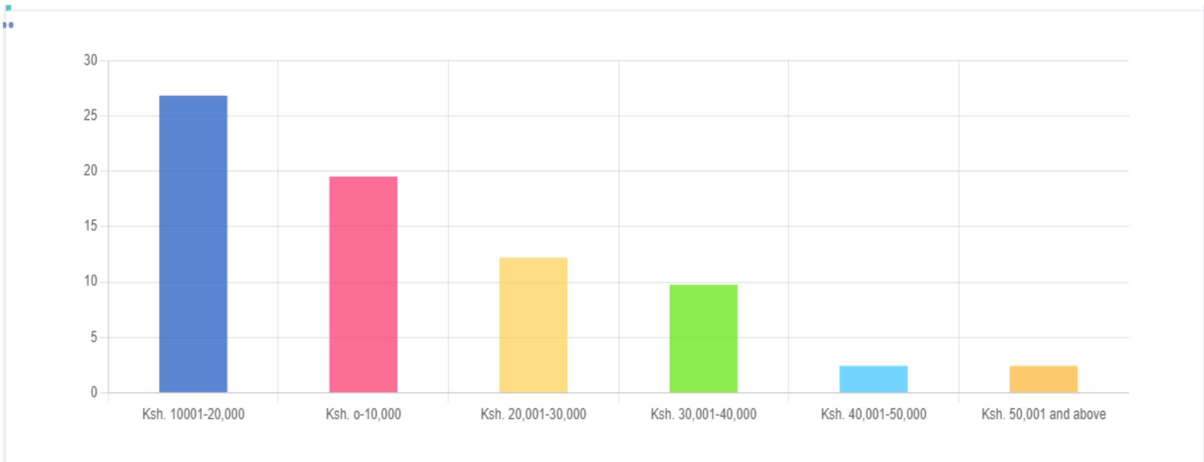
Magadi Road has emerged as a bustling commercial corridor, drawing various businesses and services to its vicinity. The road's strategic location and high traffic volume make it an attractive destination for retail outlets, restaurants, and other commercial enterprises. As a result, clusters of commercial activities have flourished along the road, contributing significantly to the linear urban pattern. This concentration of businesses not only serves the immediate needs of local residents but also attracts customers from surrounding areas, further enhancing the road's importance as a commercial hub. The graph below shows some of the major businesses thrilling along Magadi road.



Residential Settlements

The accessibility provided by Magadi Road has also facilitated the establishment of residential neighbourhoods and housing developments along its route. The convenience of transportation for commuting to work, accessing amenities, and connecting with other parts of the city makes living along Magadi Road highly desirable. As a result, residential settlements have flourished, gradually shaping the linear urban pattern observed in Ongata Rongai. This residential influx has not only expanded the local population but has also contributed to the area's social and cultural diversity.

Affordability of rent also act as a major factor that has attracted people to settle along Magadi road as the table below explain. Most residential along the road are rental while as you get in people are owning the lands they are living on with high individual density buildings.



Land Use Acceleration

Transport infrastructure along Magadi Road has spurred land use intensification, particularly in areas adjacent to the road. The high accessibility and visibility offered by the road make it an attractive location for dense residential and commercial developments. Developers capitalize on the increased land values and demand for space by maximizing land use through high-rise buildings and mixed-use developments. This intensification of land use along Magadi Road has been instrumental in shaping the linear urban pattern, characterized by pockets of high-density development along the corridor.

Transport-Oriented Development (TOD)

Magadi Road has become a focal point for transport-oriented development, guided by principles that promote mixed land uses, higher densities, and pedestrian-friendly environments. The road's importance as a transportation corridor has prompted planners and developers to envision Ongata Rongai's growth around transit nodes and along key transportation routes. This approach has led to the creation of vibrant, walkable neighbourhoods and commercial districts centred on Magadi Road, reinforcing the linear urban pattern and fostering sustainable urban development practices. Bus station like easy coach has developed along the Magadi road due to its present.

Infrastructure Investment Influence

Government investment in transport infrastructure along Magadi Road has had a profound impact on the area's urban form. Infrastructure improvements, such as road upgrades which is ongoing, public transportation systems, and pedestrian facilities, have enhanced accessibility and connectivity, attracting further development and investment along the corridor. These investments have been instrumental in shaping the linear urban pattern by catalysing growth and facilitating the efficient movement of people and goods within Ongata Rongai and its surrounding areas.

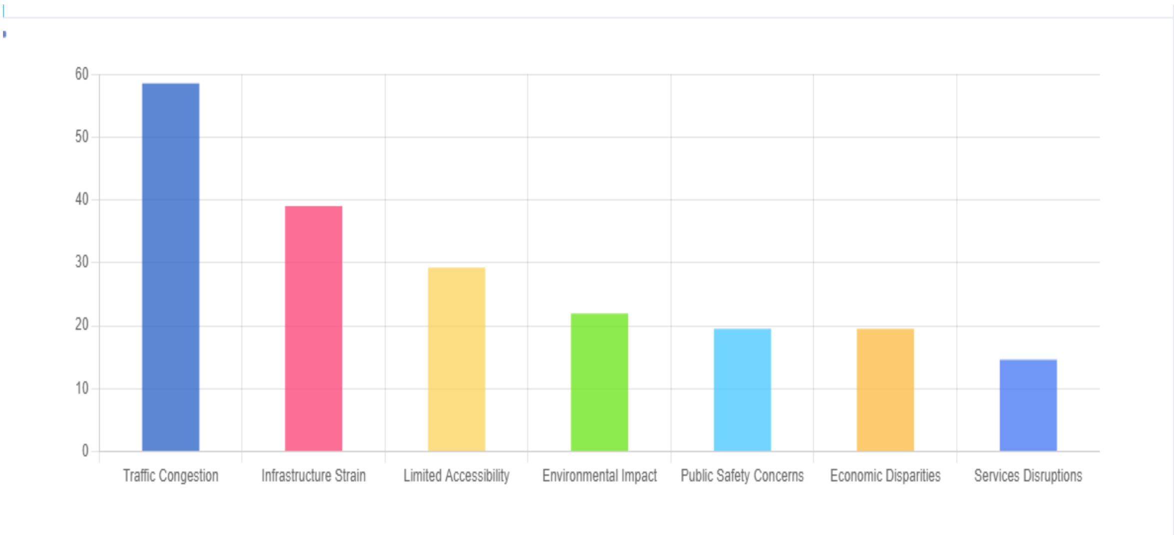
These roles demonstrate how transport infrastructure, particularly Magadi Road, has played an essential role in shaping the linear urban pattern observed in Ongata Rongai.

5.3.2To examine the effects of the linear pattern on provision of services and utilities

The effects of the linear pattern maybe positive or negative depending on the outcomes result as presented below. Although the linear urban pattern has the positive effects, they always come with burdens which have to be continuingly monitored for the benefits of the residents. Some of the negative impacts are as follow;

Negative Effects

The table below summarize some of the major challenges affecting Ongata Rongai due to its linear pattern of growth.



Traffic Congestion

Despite the benefits of concentrated development, the linear urban pattern along Magadi Road has led to traffic congestion, especially during peak hours. The concentration of businesses, residential areas, and commercial activities along the corridor increases vehicular traffic, resulting in delays, congestion, and reduced efficiency in transportation as open air market encroached the road. Traffic congestion not only affects residents' daily commutes but also poses challenges for emergency services and public transportation systems.

The above image indicates the traffic congestion experience in Rongai towards late hours of the day

Infrastructure Strain

The linear pattern of urban development has strain existing infrastructure and utilities, because they were not designed to accommodate the concentrated population density along the corridor. Increased demand for water, electricity, sanitation services, and road maintenance has exceed the capacity of existing infrastructure, leading to service disruptions, deteriorating infrastructure conditions, and increased operational costs for service providers.

Unequal Access to Services

While residents living along Magadi Road benefit from convenient access to services and amenities, those residing in peripheral areas or offshoots of the corridor have faced challenges in accessing essential services. Disparities in service provision have arisen, leading to unequal access to healthcare, education, retail, and recreational facilities within the broader Ongata Rongai area. These unequal accesses to services have exacerbated social and economic inequalities among residents.

Environmental Degradation

The concentrations of urban development along Magadi Road have contribute to environmental degradation, including air and noise pollution, as well as the loss of green spaces. Increased vehicular traffic, industrial activities, and commercial establishments along the corridor generate pollutants, negatively impacting air quality and noise levels in the surrounding areas. Additionally, the conversion of natural landscapes into built environments reduces biodiversity and diminishes the aesthetic value of the landscape.

Lack of waste management system in Ongata Rongai has led to careless disposal of waste along the road enduring even the drainage system as show by the image above

Urban Sprawl

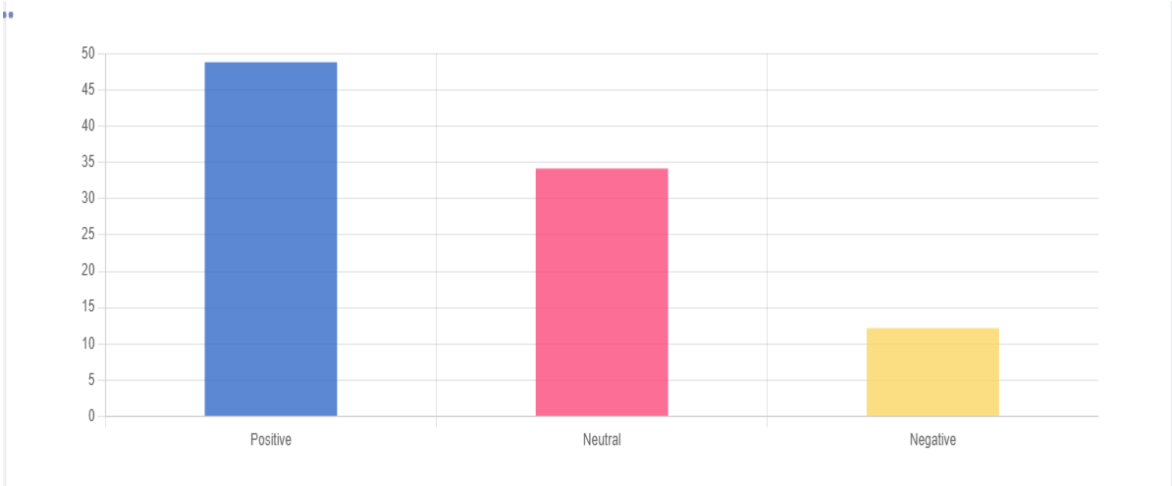
The linear pattern of urban development has encourage urban sprawl along Magadi Road, leading to inefficient land use, encroachment on agricultural land and roads, and the depletion of natural resources. The expansion of residential and commercial developments into peripheral areas have resulted in fragmented land use patterns, loss of open spaces, and increased pressure on infrastructure and ecosystems. Urban sprawl poses challenges for sustainable development, as it contributes to land degradation, habitat loss, and the degradation of natural habitats.

Social Isolation

The concentrations of urban development along the linear corridor have resulted in social isolation for residents living in offshoots or peripheral areas. These residents feel disconnected from the main urban centre and its amenities, leading to feelings of social isolation and exclusion. Lack of access to essential services, limited social interaction opportunities, and inadequate transportation options have further exacerbate social isolation among vulnerable populations, including the elderly, low-income households, and individuals with disabilities.

5.3.3 To examine the socioeconomic outcomes of the linear pattern.

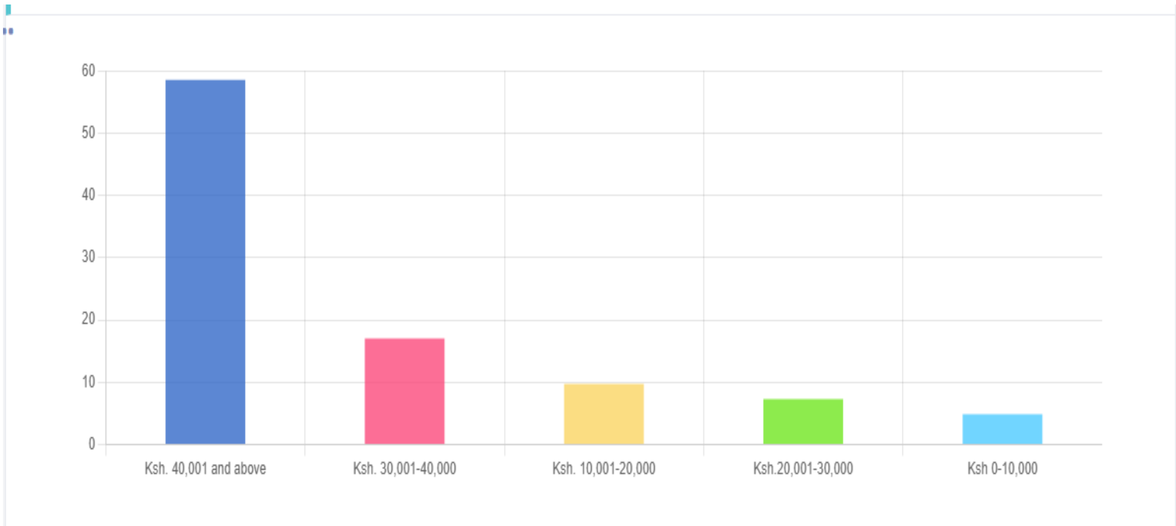
The residents managed to give their opinion on how they perceive the socioeconomic outcomes of the linear pattern and it was very clear that they feel it has improved while others feel it has remain the same as shown in the graph below.



Negative Socioeconomic Outcomes

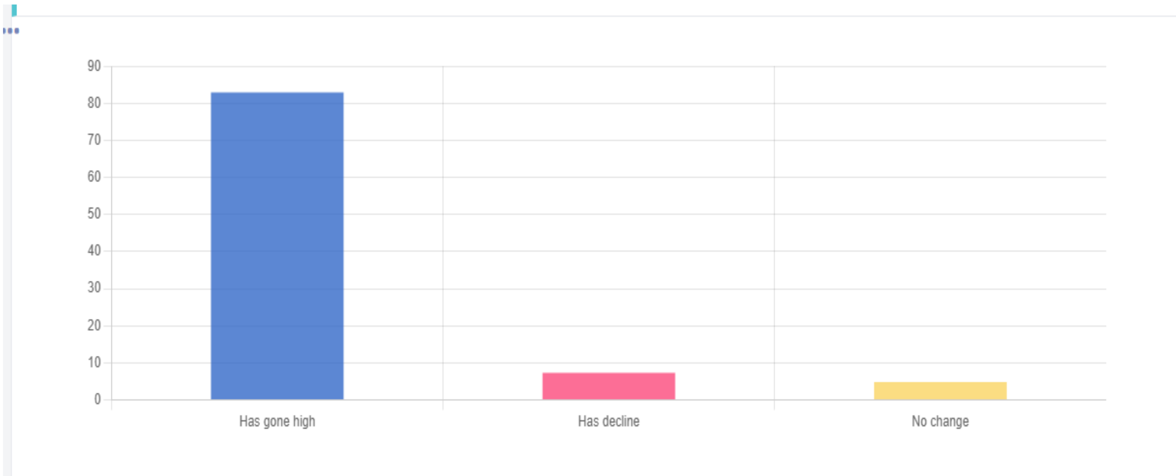
Income Inequality

The linear pattern growths along Magadi Road have exacerbated income inequality within the community. While some residents benefit from increased economic opportunities and property values, others, particularly low-income households, has face challenges in accessing affordable housing, employment, and essential services, widening socioeconomic disparities within the area. The graph shows that there are those residents with high income, medium and even lowest which can be as a result of developing other areas of Ongata Rongai while ignoring others.



Displacement of Informal Settlements

Rapid urbanization along Magadi Road has resulted in the displacement of informal settlements and vulnerable communities residing in the vicinity. As property values rise and land becomes more valuable, informal settlers may be forced to relocate to peripheral areas with limited access to services and livelihood opportunities, exacerbating poverty and social marginalization.



Redevelopment Pressures

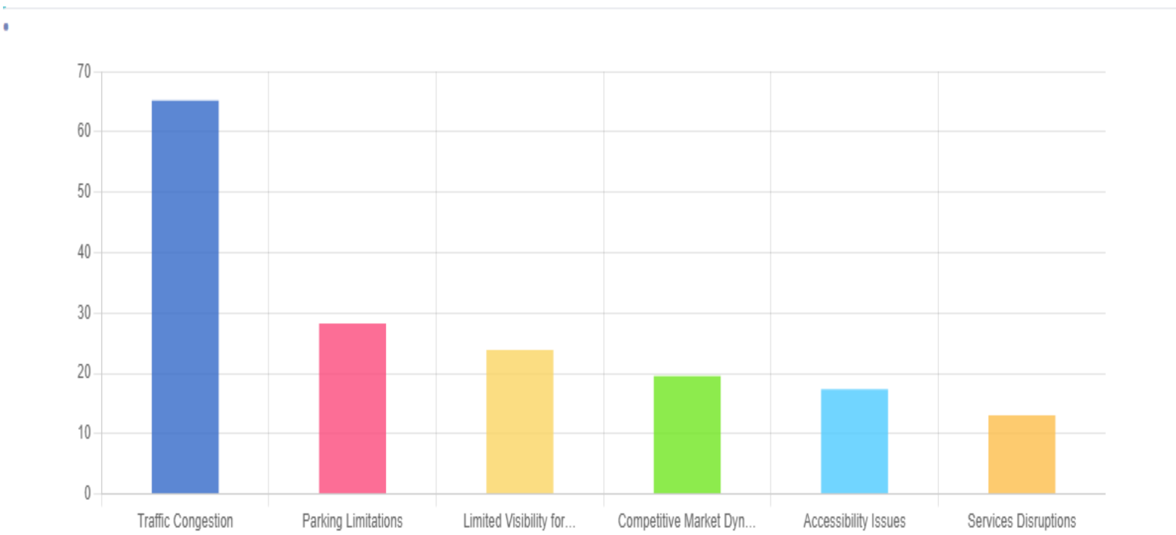
The concentrations of development along Magadi Road have led to redevelopment pressures, particularly in older residential neighbourhoods and low-income areas. Rising property values and rents has displaced long-term residents, small businesses, and cultural institutions, transforming the socio-economic fabric of the community and eroding its unique character and identity.

Strain on Social Infrastructure

The linear pattern growth places strain on social infrastructure and public services, including healthcare facilities, educational institutions, and transportation systems. Increased population density and demand for services have overwhelm existing infrastructure, leading to overcrowding, longer wait times, and reduced service quality, particularly for marginalized populations with limited access to resources and support networks.

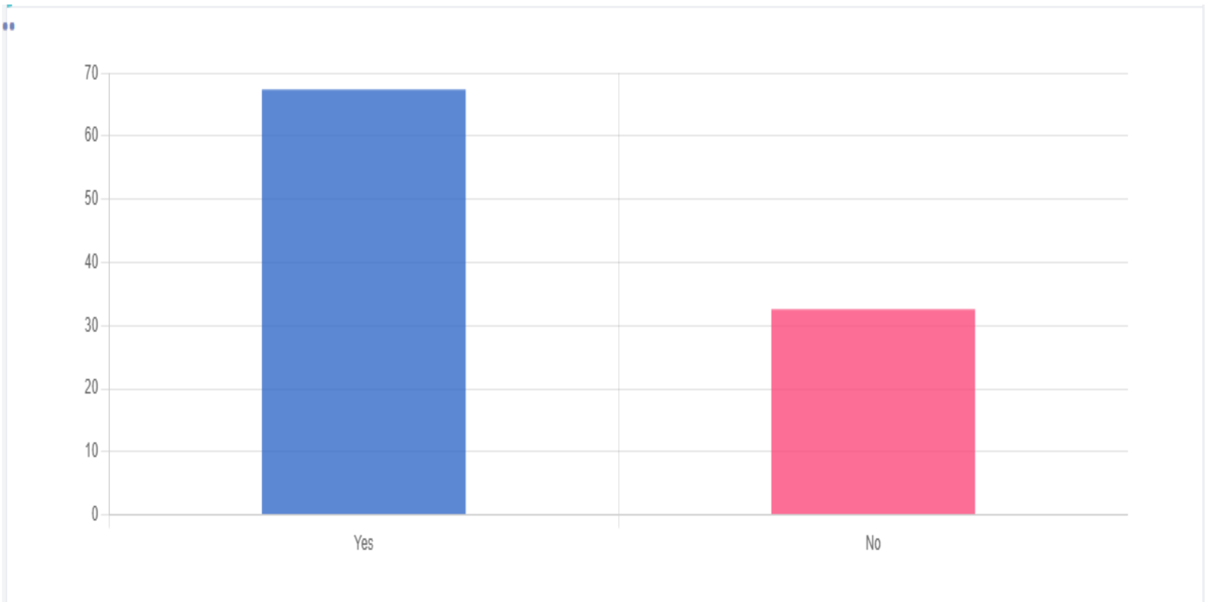
Lack of parking spaces and corridors for intersecting, privet car owners branching to their various homes is also a major problem facing the road.

Some of the prolem facing the business activity are summarize on the table below



5.3.4 To provide mitigation strategies to address the challenges arising from the linear pattern.

Development control are the process of ensuring that development applications comply with the policy guidelines, physical planning standards, approved physical development plans, local authority by laws and other relevant statutes. Before taking this survey, were you familiar with development control laws? The graph below indicates that about 70 percent of the public are not familiar with guidelines which control some of these developments. Hence there is need to train the public on the policies and regulation guiding development in the County of Kajiado.



According to Physical Planner Zakimba of Kajido County, Kajiado County has recognized the challenges associated with the linear growth of Ongata Rongai along Magadi Road and is actively pursuing various plans to address these issues. These include;

Comprehensive Urban Planning: Kajiado County is developing comprehensive urban planning frameworks that aim to guide the sustainable growth and development of Ongata Rongai. These plans include zoning regulations, land use policies, and development standards that promote compact, mixed-use development, and discourage urban sprawl along the linear corridor of Magadi Road.

Infrastructure Investments: The county government is prioritizing infrastructure investments to improve transportation networks, utilities, and social amenities along Magadi Road and its surrounding areas. This includes road upgrades which already in progress, public transportation improvements, water and sanitation projects, and the expansion of healthcare and educational facilities to accommodate the needs of a growing population.

Transit-Oriented Development (TOD): Kajiado County is promoting transit-oriented development principles to create vibrant, walkable neighbourhoods centred on public transportation hubs along Magadi Road. This approach encourages higher-density development, mixed land uses, and pedestrian-friendly environments, reducing reliance on private vehicles and enhancing mobility options for residents.

The residents also managed to give their opinions on the areas they would like to see improvement including;

1. Improvement of waste management to reduce pollution and improve public health
2. Development of public transportation would reduce traffic congestion and air pollution.
3. Investing in expanding water and sewer networks would improve access to clean water and sanitation
4. Improve coordination between government agencies to ensure they are working together to address the challenges faced by residents of Magadi road.
5. By proposing market centres to reduce markets on the road.
6. Invest in education and health care to improve the quality of life of residents and make them more productive members of the society.
7. Empower the communities, this would give residents a greater say in how their community is developed and managed.
8. Installations of traffic lights for controlled movement, construction of interlink roads for increased accessibility and reduced traffic congestion.

The mitigation is summarize on the table below

Challenges	**Mitigation**
Traffic jam	• Encourage use of public transport and designating there specific stages, plan and open corridors intersecting the residential and commercial areas. • Expansion of the road to two dual way, installation of street lights and signs.
Businesses encroaching roads and road reserves	• Awareness Campaigns to the public around the Magadi road • Regulatory Enforcement to enforce existing regulations regarding road encroachment. • Zoning Regulations by Reviewing and enforcing zoning regulations to ensure that commercial

	establishments are situated in designated areas away from roads and road reserves.
Poor waste management	• Promote Recycling and Composting to reduce the amount of waste sent to landfills. • Waste Collection Services ensure reliable and efficient waste collection services that cover all areas of the community on a regular basis.
Informal businesses along the road	• Identify and designate specific areas or markets where informal businesses can operate legally and safely.
Lack of parking facilities	• Implement parking management strategies to optimize the use of existing parking spaces and reduce congestion. This may include measures such as time limits, pricing mechanisms, and permit systems to discourage long-term parking and encourage turnover. • Promote multi-modal transportation options such as walking, cycling, and public transit to reduce reliance on private vehicles and alleviate pressure on parking facilities.
Poor management of sewer and drainage system	• Prioritize regular maintenance and repairs of sewer and drainage infrastructure to address immediate issues and prevent further deterioration. This may involve clearing blockages, repairing damaged pipes, and upgrading outdated infrastructure
Urban sprawl	• Promote transit-oriented development around public transportation corridors and hubs to encourage compact, walkable communities with access to transit options. • Establish urban growth boundaries or greenbelts to limit the outward expansion of urban development and protect valuable agricultural land, natural habitats, and open spaces. • Preserve and enhance open spaces, parks, and green corridors within urban areas to provide recreational

	opportunities, improve air quality, and mitigate the heat island effect.

CHAPTER SIX: CONCLUSIONS AND RECOMMENDATIONS

6.0 Introduction

Chapter six of this report focuses on drawing conclusions and providing recommendations based on the findings and analysis presented in the preceding chapters. Throughout this study, various aspects of have been examined. As we conclude this report, it is imperative to synthesize the findings into actionable insights and propose recommendations that can inform decision-making and guide future actions in developing Ongata Rongai.

6.1 Summary and Conclusions

Transport infrastructure, particularly Magadi Road, emerges as a critical factor in shaping the linear urban pattern observed along its route. The layout and accessibility afforded by Magadi Road have attracted significant commercial and residential developments, leading to a visible concentration of activities along this transportation corridor. This phenomenon underscores the basic relationship between transportation infrastructure and land use decisions, with Magadi Road serving as a primary axis of urban development. The linear pattern is not merely a consequence of spatial coincidence but rather a result of deliberate interactions between transport infrastructure planning and urban growth dynamics.

The effects of the linear pattern on the provision of services and utilities present a mixed picture. On one hand, the concentration of businesses and services along Magadi Road enhances accessibility for residents and businesses situated within the linear corridor. However, the heightened population density worsens challenges in service provision. Strains on utilities such as water and electricity supply, coupled with issues of congestion and inadequate waste management, highlight the necessity for careful planning and infrastructure development to sustain the demands of the linear urban pattern.

The socioeconomic outcomes of the linear pattern along Magadi Road are profound and multifaceted. Economic development flourishes due to the concentration of commercial activities and employment opportunities along the corridor. However, accompanying this prosperity are social disparities, as evidenced by variations in property values, income levels, and community cohesion within the linear pattern. The spatial arrangement of economic activities along Magadi Road influences social dynamics, underscoring the need for equitable development policies to mitigate disparities and foster inclusive growth.

In addressing the challenges arising from the linear pattern, a complicated approach is essential. Transportation planning measures, such as improving public transit and enhancing pedestrian infrastructure, can mitigate congestion and reduce reliance on private vehicles.

Spatial planning interventions, including mixed-use development promotion and optimized land use patterns, hold promise in enhancing the efficiency and sustainability of the linear corridor. Furthermore, investments in infrastructure upgrades and service delivery improvements are imperative to meet the burgeoning demands generated by the linear urban pattern. By adopting targeted mitigation strategies, policymakers can harness the benefits of the linear pattern while mitigating its negative repercussions, paving the way for sustainable and inclusive urban development along Magadi Road and similar transportation corridors.

In conclusion, the analysis reveals that road infrastructure, particularly Magadi Road, significantly influences the spatial pattern and socioeconomic dynamics of the urban area. While the linear urban pattern offers advantages in terms of accessibility and economic development, it also presents challenges related to infrastructure strain and social disparities.

Addressing these challenges requires a holistic approach that integrates transportation planning and infrastructure development. By implementing targeted mitigation strategies, policymakers can maximize the benefits of the linear pattern while minimizing its negative impacts, ultimately fostering sustainable and inclusive urban development along Magadi Road and similar transportation corridors.

6.2 Recommendations

Recommendation strategies to address the challenges arising from the linear pattern along Magadi Road in Ongata Rongai are essential for sustainable development. This includes;

Promote Land Use Planning-Implement land use planning policies that encourage mixed-use development and compact urban form along Magadi Road. This approach optimizes land use, and fosters efficient utilization of resources while promoting accessibility to services and amenities.

Affordable Housing Policies-Develop and implement affordable housing policies to address the housing needs of low-income residents affected by rising property values and gentrification pressures. This may include subsidies, incentives for affordable housing developments, and inclusionary zoning requirements to ensure the provision of housing options for diverse income groups.

Public Transportation Improvements-Invest in public transportation infrastructure and services along Magadi Road to alleviate traffic congestion, reduce reliance on private vehicles, and improve accessibility for residents. This may involve expanding bus routes, introducing dedicated lanes for buses, and enhancing pedestrian and cycling infrastructure to promote sustainable mobility options. The imagery below from Facebook shows the impotence of public transport.

Social Infrastructure Investments- Increase investment in social infrastructure, including healthcare facilities, schools, and community centres, to accommodate the needs of a growing population along Magadi Road. This entails expanding existing facilities, building new ones, and ensuring equitable access to services for all residents, particularly those in underserved areas.

Green Spaces Preservation-Protect and enhance green spaces and natural ecosystems along Magadi Road to mitigate environmental degradation and promote community health and well-being. This may involve implementing green infrastructure solutions, such as parks, urban forests, and green corridors, to enhance biodiversity, reduce pollution, and provide recreational opportunities for residents.

Community Engagement and Participation-Foster community engagement and participation in decision-making processes related to urban development along Magadi Road.

This ensures that the needs and priorities of local residents are considered in planning and implementation efforts, promoting social cohesion, inclusivity, and ownership of development initiatives.

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APPENDIX

Key Informant Interview
Name of Researcher:
Contact:

Institution: University of Nairobi

Department: Urban and Regional Planning
Program: Degree

Topic: An Analysis of the Impact of Magadi Road on Urban Spatial Pattern and provision of services. A Case of Magadi Road Ongata Rongai Town

Confidential: The information provided under this survey shall be used for this research study only and not for any other purpose.

Respondent's name:
Position:
Contact:
Gender:

1. You are a representative of(department)?
2. Which land use types are found within the larger Ongata Rongai Town?
3. Which activities have increased along the Magadi Road in the last 10 years?
4. In your opinion, what attracts people to live and invest in Rongai Town along Magadi road?
5. What challenges do you face when enforcing development control and land use zoning laws in Ongata Rongai Town as a planner?
6. In your opinion, are the residents of Rongai town familiar with the development control regulation? Why? Explain your answer.
7. In your opinion, what measures should be taken to ensure compliance with the development control guidelines?

8. Do you think the available infrastructure is sufficient in quality and quantity with the rate of development going on in Rongai town?
9. What are some of the measures in place to control and guide the undesirable linear development pattern along Magadi road as the County Government of Kajiado?

Thank you for the responds.

AN ANALYSIS OF THE IMPACT OF MAGADI ROAD ON URBAN'S SPATIAL PATTERN ONGATA RONGAI TOWN

THE UNIVERSITY OF NAIROBI STUDENT RESEARCH PROJECT BUSINESS QUESTIONNAIRE

1. What is your gender?

○ Male ○ Female

2. What is your age?

○ 18-25 ○ 26-35 ○ 36-45 ○ 46-55 ○ 56 and above

3. What is the nature of your business?

○ Restaurant and takeaways ○ Recreational facility ○ Retail and wholesale (shops, supermarkets) ○ Financial services(mpesa, banks) ○ Engineering, building and construction material ○ Open air market supplier ○ Education ○ Electronics supplier ○ Pharmacy ○ Clothing shops

4. For how long has your business been operating here?5. Average monthly income from the business

○ Ksh. 0-50,000 ○ Ksh. 50,001-100,000 ○ Ksh. 100,001-200,000 ○ Ksh. 200.001-500,000 ○ Ksh. 500,001 and above

6. What made you establish your business in this area? You can tick more than 1

☐ Availabity of land ☐ Market opportunity for my product ☐ Availability of access e.g road infrastructure ☐ Access to facilities ☐ Proximity to my home/house ☐ Ease of doing business (no barriers to the market) ☐ Others (specify)

7. How does the Magadi road contribute to your business income?

☐ More human traffic into the business income?☐ The business is more visible from the road☐ Ease access☐ Economic Development☐ Other (specify)

8. Which factors have favoured the growth and development of the Magadi road neighbourhoods? You can tick more than one

☐ Strategic Location☐ Transportation infrastructure☐ High Human Traffic☐ Increase of the number of businesses☐ Availability of Security☐ Affordable land☐ Community Services and social amenities e.g schools, hospital☐ Other (specify)

9. How has the linear urban pattern affected the availability and reliability of essential services for your business (e.g., water, electricity, waste management)?

○ Positively○ Neutral○ Negatively

10. What challenges do you think arise from the linear urban pattern along Magadi Road that affects businesses? You can choose more than one

☐ Limited Visibility for Businesses☐ Traffic Congestion☐ Parking Limitations☐ Accessibility Issues☐ Services Disruptions☐ Competitive Market Dynamics☐ Other (specify)

11. Are there any suggested mitigation strategies or interventions that you think could address these challenges?

12.Development control are the process of ensuring that development applications comply with the policy guidelines, physical planning standards, approved physical development plans, local authority by laws and other relevant statutes. Before taking this survey, were you familiar with development control laws?

○ Yes◉ No

Thank you for your valuable input. Your responses will contribute significantly to understanding the impact of road infrastructure on businesses and the overall urban spatial pattern along Magadi Road.

AN ANALYSIS OF THE IMPACT OF MAGADI ROAD ON URBAN SPATIAL PATTERN AND PROVISION OF SERVICES

THE UNIVERSITY OF NAIROBI STUDENT FIELD WORK SURVEY, HOUSEHOLD QUESTIONNAIRE

1. What is your gender?

Male Female

2. What is your age?

18-25 26-35 36-45 46-55 56 and above

3. How many years have you been residing in rongai?4. Are you employed?

Yes No

5. If yes, what is your occupation? 6. Where do you work? *e.g Rongai, Nairobi, Machakos, Makueni* 7. What made you to choose this location as your area of resident?

Proximity to work place Affordable rent Security Access to social amenities (schools, hospital) Good road network Recreational areas other(specify)

8. If other specify 9. Do you own the land you are residing on?

Yes No

10. If no, how much rent do you pay per month?

Ksh. o-10,000 Ksh. 10001-20,000 Ksh. 20,001-30,000 Ksh. 30,001-40,000 Ksh. 40,001-50,000 Ksh. 50,001 and above

11. Average monthly income of the household head

Ksh 0-10,000 Ksh. 10,001-20,000 Ksh.20,001-30,000 Ksh. 30,001-40,000 Ksh. 40,001 and above

12. Have you observed a linear urban pattern along Magadi Road?

Yes No

13. How has the linear urban pattern along Magadi Road affected the provision of essential services in your area (e.g., water supply, electricity, waste management)?

Improved No change Declined

14. How do you perceive the socioeconomic outcomes of the linear pattern on businesses and economic activities in Ongata Rongai?

○ Positive ○ Neutral ○ Negative

15. In your opinion, how has the linear pattern influenced property values and housing markets in the area?

○ Has gone high ○ No change ○ Has decline

17. What challenges do you think arise from the linear urban pattern along Magadi Road?

☐ Traffic Congestion ☐ Limited Accessibility ☐ Services Disruptions ☐ Economic Disparities ☐ Environmental Impact ☐ Infrastructure Strain ☐ Public Safety Concerns ☐ Any other specify

18. Are there any suggested mitigation strategies or interventions you think could address these challenges?

Thank you for your valuable input. Your responses will contribute significantly to the understanding of the impact of road infrastructure on the urban spatial pattern along Magadi Road.